

Performance Analysis

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Detailed analysis of your performance in the AWS Certified Solutions Architect Associate - SAA C03 exam.

Answer Distribution

0

Correct

0

Incorrect

65

Unattempted

Domain Performance

Security	0%
Networking	0%
Compute	0%
Monitoring	0%
Storage	0%
Database	0%
Serverless	0%
Automation	0%

Final Score

**0/65**

Passing score: 48

Time Taken

**130m 0s**

Total time: 130 minutes



Question Analysis

✓ Correct	0
✗ Incorrect	0
⊖ Unattempted	65
🔖 Reviewed	0

Question-wise Analysis

Expand All

Collapse All

Question 1

Not Attempted

Security



A company hosts an application on Amazon EC2 instances behind an Application Load Balancer (ALB). The application serves clients in 20,000+ retail storefront locations globally, communicating over HTTPS on port 443 via the public internet. Each location registers its allocated IP address from its local ISP. The security team wants to restrict access to the application endpoint to only these registered IP addresses. What is the MOST efficient and scalable solution for a solutions architect to implement?

- ✓ Associate an AWS WAF web ACL with the ALB. Use IP rule sets in the WAF to filter traffic, updating the rules with the registered IP addresses.

AWS WAF allows you to create rule sets based on IP addresses to filter traffic. It's designed for web application protection and integrates seamlessly with ALBs. The scalability of WAF supports a large number of IP addresses.

- ✗ Deploy AWS Firewall Manager to manage AWS Firewall rules. Configure firewall rules to restrict traffic to the ALB and modify them to include the registered IP addresses.

AWS Firewall Manager is primarily for managing firewalls across multiple accounts and resources. While it can manage WAF rules, it's not the most direct approach for this specific requirement.

- ✗ Store the IP addresses in an Amazon DynamoDB table. Configure an AWS Lambda authorization function on the ALB to validate that incoming requests are from the registered IP addresses.

While technically feasible, using Lambda for IP filtering adds latency and complexity. It's less efficient than using WAF's built-in IP filtering capabilities, which is better suited to handle this kind of task.

- ✗ Configure the network ACL on the subnet containing the ALB's public interface. Update the ingress rules with entries for each registered IP address.

Network ACLs have limitations on the number of rules, making them unsuitable for managing a large number of IP addresses (20,000+). Additionally, NACLs operate at the subnet level and are not specific to the application, and therefore not the ideal solution.

Overall Explanation:

AWS WAF is the most appropriate choice. WAF can be associated with an ALB and used to filter traffic based on IP address. It is designed for web application protection and scales well to handle a large number of IP addresses (20,000+). Other options are either less efficient (Lambda) or not designed for the scale and specificity required (Network ACLs, Firewall Manager).

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Question 2

Not Attempted

Security



A company is building a data analysis platform on AWS using AWS Lake Formation. The platform ingests data from various sources like Amazon S3 and Amazon RDS. To ensure data

security, the company must prevent access to sensitive information within the data. What solution achieves this with the LEAST operational overhead?

- ☐ Create an IAM role that includes permissions to access Lake Formation tables.
IAM roles provide user-level permissions but do not offer row-level or cell-level security.
- ☒ Create data filters to implement row-level security and cell-level security.
Lake Formation data filters allow restricting access to rows or cells in data tables based on conditions, effectively preventing access to sensitive data without additional coding or infrastructure.
- ☐ Create an AWS Lambda function that removes sensitive information before Lake Formation ingests the data.
Using a Lambda function for pre-processing requires ongoing development, maintenance, and operational overhead.
- ☐ Create an AWS Lambda function that periodically queries and removes sensitive information from Lake Formation tables.
Using a Lambda function to periodically purge data incurs more operational overhead and complexity than built-in data filters.

Overall Explanation:

Data filters are the most efficient solution. They are a built-in feature of Lake Formation designed to implement row-level and cell-level security, preventing access to sensitive information with minimal operational overhead. The other options involve more complex solutions and require more resources to manage and maintain.

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Question 3

Not Attempted

Security



A company is developing a two-tier web application on AWS. The company's developers have deployed the application on an Amazon EC2 instance that connects directly to a backend Amazon RDS database. The company must not hardcode database credentials in the application. The company must also implement a solution to automatically rotate the database credentials on a regular basis. Which solution will meet these requirements with the LEAST operational overhead?

- ⊗ Store the database credentials in the instance metadata. Use Amazon EventBridge (Amazon CloudWatch Events) rules to run a scheduled AWS Lambda function that updates the RDS credentials and instance metadata at the same time.
Storing credentials in instance metadata and using Lambda for rotation is complex and has high operational overhead.
- ⊗ Store the database credentials in a configuration file in an encrypted Amazon S3 bucket. Use Amazon EventBridge (Amazon CloudWatch Events) rules to run a scheduled AWS Lambda function that updates the RDS credentials and the credentials in the configuration file at the same time. Use S3 Versioning to ensure the ability to fall back to previous values.
Storing credentials in S3 and using Lambda for rotation is complex, and managing versions adds operational overhead.
- ✓ Store the database credentials as a secret in AWS Secrets Manager. Turn on automatic rotation for the secret. Attach the required permission to the EC2 role to grant access to the secret.
Secrets Manager provides automatic rotation with minimal overhead and EC2 roles can be used to securely access the secret. This is the most efficient solution.
- ⊗ Store the database credentials as encrypted parameters in AWS Systems Manager Parameter Store. Turn on automatic rotation for the encrypted parameters. Attach the required permission to the EC2 role to grant access to the encrypted parameters.
Parameter Store does not have automatic rotation, so this is not a viable solution.

Overall Explanation:

AWS Secrets Manager is the best solution because it is designed for managing secrets, allows for automatic rotation, and integrates seamlessly with EC2 instance roles for secure access with the least operational overhead. Parameter store does not support autorotation, and using Lambda to update instance metadata or S3 is more complex and creates overhead. AWS Secrets Manager handles the credential updates automatically. Attaching the necessary permissions to the EC2 role allows the application to securely access the secret.

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A company deploys Amazon EC2 instances that run in a VPC. The EC2 instances load source data into Amazon S3 buckets so that the data can be processed in the future. According to compliance laws, the data must not be transmitted over the public internet. Servers in the company's on-premises data center will consume the output from an application that runs on the EC2 instances. Which solution will meet these requirements?

- ☐ Deploy an interface VPC endpoint for Amazon EC2. Create an AWS Site-to-Site VPN connection between the company and the VPC.

This option is incorrect because there's no such thing as an interface VPC endpoint for EC2. Also, Site-to-Site VPN connections use the public internet.

- ☒ Deploy a gateway VPC endpoint for Amazon S3. Set up an AWS Direct Connect connection between the on-premises network and the VPC.

This option is correct. A gateway VPC endpoint for S3 allows EC2 instances to access S3 without using the public internet. AWS Direct Connect provides a private connection between the on-premises data center and the VPC, ensuring no data is transmitted over the public internet.

- ☐ Set up an AWS Transit Gateway connection from the VPC to the S3 buckets. Create an AWS Site-to-Site VPN connection between the company and the VPC.

This option is incorrect because Transit Gateway is not typically used for direct VPC to S3 communication in this scenario and Site-to-Site VPN uses the public internet.

- ☐ Set up proxy EC2 instances that have routes to NAT gateways. Configure the proxy EC2 instances to fetch S3 data and feed the application instances.

This option is incorrect because NAT gateways use the public internet.

Overall Explanation:

The correct solution is to deploy a gateway VPC endpoint for Amazon S3 and set up an AWS Direct Connect connection between the on-premises network and the VPC. The gateway VPC endpoint allows EC2 instances within the VPC to access Amazon S3 without traversing the public internet. The AWS Direct Connect connection provides a private, dedicated network connection between the on-premises data center and the VPC, ensuring that the data transferred between the on-premises servers and the EC2 instances does not traverse the public internet, thus meeting the compliance requirements.

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Question 5

Not Attempted

Compute



A company has an application with a REST-based interface that allows data to be received in near-real time from a third-party vendor. Once received, the application processes and stores the data for further analysis. The application is running on Amazon EC2 instances. The third-party vendor has received many 503 Service Unavailable Errors when sending data to the application. When the data volume spikes, the compute capacity reaches its maximum limit, and the application is unable to process all requests. Which design should a solutions architect recommend to provide a more scalable solution?

- ☒ Use Amazon Kinesis Data Streams to ingest the data. Process the data using AWS Lambda functions.
Kinesis Data Streams provides auto-scaling stream for large streaming data ingestion. AWS Lambda can process and store data in a scalable serverless manner, avoiding EC2 capacity limits.
- ☐ Use Amazon API Gateway on top of the existing application. Create a usage plan with a quota limit for the third-party vendor.
API Gateway adds API management capabilities but does not improve the underlying scalability of the EC2 application.
- ☐ Use Amazon Simple Notification Service (Amazon SNS) to ingest the data. Put the EC2 instances in an Auto Scaling group behind an Application Load Balancer.
SNS is for event publishing/notifications, not large scale data ingestion.
- ☐ Repackage the application as a container. Deploy the application using Amazon Elastic Container Service (Amazon ECS) using the EC2 launch type with an Auto Scaling group.
ECS using EC2 still relies on EC2 capacity and doesn't fully address scalability issues compared to a serverless approach.

Overall Explanation:

The most scalable solution is to use Amazon Kinesis Data Streams to ingest the data in near-real time and then process the data using AWS Lambda functions. Kinesis handles the high volume of streaming data, and Lambda provides a serverless environment for processing, avoiding the limitations of EC2 capacity. This serverless approach ensures the application can scale effectively during data volume spikes. Therefore, Option A is the correct answer.

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Question 6

Not Attempted

Networking

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A company hosts an application on Amazon EC2 instances within a private subnet. This application is required to process sensitive data located in an Amazon S3 bucket. To comply with security policies, the application **MUST NOT** utilize the internet to access the S3 bucket. Which solution satisfies these requirements?

- ❌ **Configure an internet gateway.** Update the S3 bucket policy to allow access from the internet gateway. Update the application to use the new internet gateway.

This option exposes the S3 bucket to the internet, which violates the requirement of not using the internet for S3 access. An internet gateway provides internet access, which is the opposite of what's needed.
- ❌ **Configure a VPN connection.** Update the S3 bucket policy to allow access from the VPN connection. Update the application to use the new VPN connection.

While a VPN provides a secure connection, it's typically used for connecting on-premises networks to AWS. It doesn't inherently prevent internet usage for S3 access and adds unnecessary complexity for this scenario.
- ❌ **Configure a NAT gateway.** Update the S3 bucket policy to allow access from the NAT gateway. Update the application to use the new NAT gateway.

A NAT gateway allows instances in a private subnet to initiate outbound internet traffic, but it doesn't enable private connectivity to S3. It's used for accessing the internet, which is against the requirements.
- ✅ **Configure a VPC endpoint.** Update the S3 bucket policy to allow access from the VPC endpoint. Update the application to use the new VPC endpoint.

A VPC endpoint for S3 creates a private connection between the VPC and S3, without requiring an internet gateway, NAT device, or VPN connection. It enables the application to access S3 directly within the AWS network, fulfilling the requirement of not using the internet.

Overall Explanation:

The correct solution is to configure a VPC endpoint for S3. A VPC endpoint provides private connectivity between your VPC and S3, ensuring that traffic does not traverse the internet. By updating the S3 bucket policy to allow access from the VPC endpoint and configuring the application to use it, you meet the requirement of accessing S3 without

internet access. Options involving internet gateways, VPN connections or NAT gateways are incorrect because they rely on or facilitate internet traffic, contradicting the question's requirements.

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Question 7

Not Attempted

Security



A company uses Amazon Elastic Kubernetes Service (Amazon EKS) to run a container application. The EKS cluster stores sensitive information in Kubernetes secrets objects. The company needs to ensure that this sensitive information is encrypted at rest with the LEAST amount of operational overhead. Which solution meets these requirements?

- ☐ Use the container application to encrypt the information by using AWS Key Management Service (AWS KMS).
Encrypting secrets within the application itself introduces code complexity and increases operational overhead.
- ☒ Enable secrets encryption in the EKS cluster by using AWS Key Management Service (AWS KMS).
EKS supports encrypting Kubernetes secrets at the cluster level using KMS, providing an automated and low-overhead solution.
- ☐ Implement an AWS Lambda function to encrypt the information by using AWS Key Management Service (AWS KMS).
Using a Lambda function adds unnecessary complexity and operational overhead compared to native EKS secrets encryption.
- ☐ Use AWS Systems Manager Parameter Store to encrypt the information by using AWS Key Management Service (AWS KMS).
Parameter Store can store secrets, but it doesn't natively integrate with EKS for encrypting Kubernetes secrets objects. This approach would introduce more operational overhead.

Overall Explanation:

Enabling secrets encryption in the EKS cluster using AWS KMS is the most straightforward solution. EKS natively supports this feature, ensuring encryption at rest without requiring modifications to the application code or the introduction of additional components like

Lambda functions. Using KMS directly within the application or through Lambda or Parameter Store adds operational overhead compared to EKS's built-in encryption capabilities.

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Question 8

Not Attempted

Security



A company is designing a new multi-tier web application consisting of web and application servers on Amazon EC2 instances within Auto Scaling groups, and an Amazon RDS DB instance for data storage. The solutions architect must restrict access to the application servers so that only the web servers can access them. Which solution effectively meets these requirements?

- ☐ Deploy AWS PrivateLink in front of the application servers. Configure the network ACL to allow only the web servers to access the application servers.

AWS PrivateLink is for connecting VPCs or accessing AWS services privately, not for internal access control within the same VPC.

- ☐ Deploy a VPC endpoint in front of the application servers. Configure the security group to allow only the web servers to access the application servers.

VPC endpoints are used to provide private access to AWS services and are not designed for controlling access between EC2 instances within a VPC.

- ☐ Deploy a Network Load Balancer with a target group that contains the application servers' Auto Scaling group. Configure the network ACL to allow only the web servers to access the application servers.

While an NLB can distribute traffic, network ACLs operate at the subnet level and are less granular than security groups for instance-level access control. Also, relying solely on NACLs can lead to asymmetrical routing issues.

- ☒ Deploy an Application Load Balancer with a target group that contains the application servers' Auto Scaling group. Configure the security group to allow only the web servers to access the application servers.

This is the correct answer. An ALB provides a manageable endpoint for the application servers' ASG, and security groups allow fine-grained control over which instances (web servers) can access the application servers.

Overall Explanation:

The most effective solution is to use an Application Load Balancer (ALB) in front of the application servers' Auto Scaling group. This provides a single point of access that can be controlled using security groups. By configuring the security group associated with the application servers, you can explicitly allow traffic only from the security group associated with the web servers, effectively limiting access as required.

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Question 9

Not Attempted

Monitoring



A company runs a critical, customer-facing application on Amazon Elastic Kubernetes Service (Amazon EKS). The application has a microservices architecture. The company needs to implement a solution that collects, aggregates, and summarizes metrics and logs from the application in a centralized location. Which solution meets these requirements?

- ☐ Run the Amazon CloudWatch agent in the existing EKS cluster. View the metrics and logs in the CloudWatch console.

Running the CloudWatch agent alone may not provide the aggregation and summarization features needed for a microservices architecture.

- ☐ Run AWS App Mesh in the existing EKS cluster. View the metrics and logs in the App Mesh console.

AWS App Mesh focuses on service mesh functionality, traffic management and security, not centralized log and metric aggregation.

- ☐ Configure AWS CloudTrail to capture data events. Query CloudTrail by using Amazon OpenSearch Service.

AWS CloudTrail captures API activity and is not suitable for collecting application metrics and logs.

- ☒ Configure Amazon CloudWatch Container Insights in the existing EKS cluster. View the metrics and logs in the CloudWatch console.

Amazon CloudWatch Container Insights is specifically designed to collect, aggregate, and summarize metrics and logs from containerized applications, making it the ideal solution.

Overall Explanation:

Amazon CloudWatch Container Insights provides a comprehensive solution for collecting, aggregating, and summarizing metrics and logs from containerized applications and

microservices running on Amazon EKS. It provides visibility into resource utilization, application performance, and microservice interactions. The metrics and logs are stored in CloudWatch Logs and CloudWatch metrics for central access, and the CloudWatch console allows querying, filtering, and visualizing the data.

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Question 10

Not Attempted

Security



A company has deployed its newest product on AWS. The product runs in an Auto Scaling group behind a Network Load Balancer. The company stores the product's objects in an Amazon S3 bucket. The company recently experienced malicious attacks against its systems. The company needs a solution that continuously monitors for malicious activity in the AWS account, workloads, and access patterns to the S3 bucket. The solution must also report suspicious activity and display the information on a dashboard. Which solution will meet these requirements?

- ❌ **Configure Amazon Macie to monitor and report findings to AWS Config.**
Amazon Macie primarily focuses on discovering and protecting sensitive data. It does not provide comprehensive monitoring for malicious activities across the entire AWS account and workloads.
- ❌ **Configure Amazon Inspector to monitor and report findings to AWS CloudTrail.**
Amazon Inspector focuses on automated security assessments of applications and identifying vulnerabilities. It doesn't provide continuous monitoring for malicious activity and doesn't directly integrate with CloudTrail for reporting findings.
- ✅ **Configure Amazon GuardDuty to monitor and report findings to AWS Security Hub.**
Amazon GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior. It analyzes AWS CloudTrail, VPC Flow Logs, and DNS logs. Findings can be sent to AWS Security Hub which provides a centralized security dashboard and alerts.
- ❌ **Configure AWS Config to monitor and report findings to Amazon EventBridge.**
AWS Config monitors for resource configuration changes, not malicious activity. It doesn't provide the threat detection capabilities needed for this scenario.

Amazon GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior to protect AWS accounts, Amazon Elastic Compute Cloud (EC2) workloads, container applications, Amazon Aurora databases, and data stored in Amazon Simple Storage Service (S3). It integrates well with AWS Security Hub, providing a centralized security dashboard.

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Question 11

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Storage



A company is migrating its on-premises data center to AWS. The data center uses a storage server with a 200 GB NFS-based file system. The company needs to migrate the data to AWS without disrupting existing services. Several AWS resources must be able to access the data using the NFS protocol. Which combination of actions will meet these requirements in the MOST cost-effective way?

☐ Create an Amazon FSx for Lustre file system.

FSx for Lustre is designed for high-performance computing and not primarily for NFS-based file systems. It's not the most cost-effective option for this scenario.

☒ Create an Amazon Elastic File System (Amazon EFS) file system.

Amazon EFS is a scalable, high-performance NFS file system that can be accessed by multiple resources in AWS, meeting the requirement of using the NFS protocol.

☐ Create an Amazon S3 bucket to receive the data.

Amazon S3 is object storage, not a file system using NFS. Therefore, it does not fulfill the requirement of accessing the data using the NFS protocol.

☐ Manually use an operating system copy command to push the data into the AWS destination.

Manually copying 200 GB of data could be slow, risky, and might cause downtime. It's not an efficient approach for migration.

☒ Install an AWS DataSync agent in the on-premises data center. Use a DataSync task between the on-premises location and AWS.

AWS DataSync allows for incremental data transfer and supports migrating NFS data without service interruption. This ensures data integrity and minimizes downtime.

Overall Explanation:

Amazon EFS provides a scalable NFS file system, enabling multiple AWS resources to access the data via the NFS protocol. AWS DataSync facilitates data migration from the on-premises NFS server to EFS without service interruptions. DataSync incrementally syncs changed data making it a cost-effective migration solution compared to manual copying or using Amazon FSx for Lustre.

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Question 12

Not Attempted

Storage



A company needs to deploy Amazon FSx for Windows File Server for its Amazon EC2 instances in the us-east-1 Region. The file system will be mounted as a shared volume. The company requires a Recovery Point Objective (RPO) of 5 minutes for planned system maintenance and unplanned disruptions. The file system must be replicated to the us-west-2 Region. Furthermore, the data in the replicated file system must be protected from deletion by any user for a period of 5 years. Which solution will meet the company's requirements?

- ☐ Create an FSx for Windows File Server file system in us-east-1 that has a Single-AZ 2 deployment type. Use AWS Backup to create a daily backup plan that includes a backup rule that copies the backup to us-west-2. Configure AWS Backup Vault Lock in compliance mode for a target vault in us-west-2. Configure a minimum duration of 5 years.

Single-AZ deployments do not meet the RPO of 5 minutes for unplanned disruptions.

- ☐ Create an FSx for Windows File Server file system in us-east-1 that has a Multi-AZ deployment type. Use AWS Backup to create a daily backup plan that includes a backup rule that copies the backup to us-west-2. Configure AWS Backup Vault Lock in governance mode for a target vault in us-west-2. Configure a minimum duration of 5 years.

Governance mode for Vault Lock does not prevent deletion by all users, and would not meet the 5 year retention requirement.

- ☒ Create an FSx for Windows File Server file system in us-east-1 that has a Multi-AZ deployment type. Use AWS Backup to create a daily backup plan that includes a backup rule that copies the backup to us-west-2. Configure AWS Backup Vault Lock in

compliance mode for a target vault in us-west-2. Configure a minimum duration of 5 years.

This option meets all requirements: Multi-AZ deployment provides high availability for the required RPO, AWS Backup replicates the data to us-west-2, and Compliance mode for Vault Lock prevents deletion for the configured duration.

- ⊗ Create an FSx for Windows File Server file system in us-east-1 that has a Single-AZ 2 deployment type. Use AWS Backup to create a daily backup plan that includes a backup rule that copies the backup to us-west-2. Configure AWS Backup Vault Lock in governance mode for a target vault in us-west-2. Configure a minimum duration of 5 years.

Single-AZ deployments do not meet the RPO of 5 minutes and Governance mode for Vault Lock does not prevent deletion by all users.

Overall Explanation:

The correct answer is C. Multi-AZ deployment for FSx for Windows File Server is required to meet the RPO of 5 minutes. AWS Backup is used to copy backups to the us-west-2 region. AWS Backup Vault Lock in Compliance mode prevents deletion of the backups for the specified duration of 5 years, fulfilling the data retention requirements.

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Question 13

Not Attempted

Security



A solutions architect is designing a security solution for a company that wants to provide developers with individual AWS accounts through AWS Organizations, while also maintaining standard security controls. Because the individual developers will have AWS account root user-level access to their own accounts, the solutions architect wants to ensure that the mandatory AWS CloudTrail configuration that is applied to new developer accounts is not modified. Which action meets these requirements?

- ⊗ Create an IAM policy that prohibits changes to CloudTrail and attach it to the root user. Attaching an IAM policy to the root user within each account isn't the most effective way to centrally manage and enforce compliance across multiple accounts in an AWS Organization. SCPs are better suited for this purpose.

- ⊗ Create a new trail in CloudTrail from within the developer accounts with the organization trails option enabled.

Creating trails from within the developer accounts doesn't prevent modifications to the initially configured mandatory trail. Also, this approach doesn't centrally enforce the CloudTrail configuration.

- ✔ Create a service control policy (SCP) that prohibits changes to CloudTrail and attach it to the developer accounts.

This is the correct answer. SCPs allow you to centrally control the permissions available within member accounts in an AWS Organization. By creating an SCP that denies changes to CloudTrail and attaching it to the developer accounts, you ensure that the mandatory CloudTrail configuration cannot be modified, even by the root user in those accounts.

- ⊗ Create a service-linked role for CloudTrail with a policy condition that allows changes only from an Amazon Resource Name (ARN) in the management account.

Service-linked roles don't inherently prevent modifications by users within the developer accounts. While a policy condition can restrict changes, it doesn't provide the same level of enforcement and centralized control as an SCP.

Overall Explanation:

SCPs are designed to centrally manage permissions and enforce compliance across multiple AWS accounts within an AWS Organization. By using an SCP to restrict changes to CloudTrail, the architect ensures that the mandatory configuration remains intact across the developer accounts, regardless of their root user access.

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Question 14

Not Attempted

Security



A company is deploying a new public web application to AWS. The application will run behind an Application Load Balancer (ALB). The application needs to be encrypted at the edge with an SSL/TLS certificate that is issued by an external certificate authority (CA). The certificate must be rotated each year before the certificate expires. What should a solutions architect do to meet these requirements?

- ⊗ Use AWS Certificate Manager (ACM) to issue an SSL/TLS certificate. Apply the certificate to the ALB. Use the managed renewal feature to automatically rotate the certificate.

This is incorrect because the certificate is issued by an external CA, not ACM. ACM's managed renewal feature only works for ACM-issued certificates.

- ⊗ Use AWS Certificate Manager (ACM) to issue an SSL/TLS certificate. Import the key material from the certificate. Apply the certificate to the ALB. Use the managed renewal feature to automatically rotate the certificate.

This is incorrect because it still attempts to use ACM to issue the certificate, which contradicts the requirement of using an external CA. Also, you don't import the key material separately when importing a certificate into ACM.

- ⊗ Use AWS Certificate Manager (ACM) Private Certificate Authority to issue an SSL/TLS certificate from the root CA. Apply the certificate to the ALB. Use the managed renewal feature to automatically rotate the certificate.

This is incorrect because ACM Private CA is for issuing private certificates within an organization and doesn't integrate with external CAs.

- ✓ Use AWS Certificate Manager (ACM) to import an SSL/TLS certificate. Apply the certificate to the ALB. Use Amazon EventBridge (Amazon CloudWatch Events) to send a notification when the certificate is nearing expiration. Rotate the certificate manually.

This is the correct approach. ACM allows importing certificates from external CAs. Since ACM cannot automatically renew imported certificates, EventBridge can be used to trigger notifications for manual renewal.

Overall Explanation:

The correct solution is to import the externally issued SSL/TLS certificate into AWS Certificate Manager (ACM). ACM allows you to manage imported certificates, but it doesn't automatically renew them. To handle certificate rotation, Amazon EventBridge (formerly CloudWatch Events) should be configured to send notifications when the certificate is nearing expiration. This allows for manual rotation of the certificate before it expires, fulfilling the requirements.

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Question 15

Not Attempted

Storage



A company intends to migrate a business-critical application to the AWS Cloud. The application demands highly durable storage coupled with consistent, low-latency performance. Which storage option should the Solutions Architect recommend?

⊗ Instance store volume

Instance store volumes provide temporary block storage for EC2 instances and are not durable. Data stored on instance store volumes is lost when the instance is stopped, terminated, or fails.

⊗ Amazon ElastiCache for Memcached cluster

Amazon ElastiCache for Memcached is an in-memory data store service, suitable for caching data, and is not designed for durable, persistent storage required for a business-critical application.

✓ Provisioned IOPS SSD Amazon Elastic Block Store (Amazon EBS) volume

Provisioned IOPS SSD EBS volumes are designed for I/O-intensive workloads and offer consistent, low-latency performance with the ability to specify the required IOPS. They provide durable block storage for EC2 instances, making them a suitable choice for business-critical applications.

⊗ Throughput Optimized HDD Amazon Elastic Block Store (Amazon EBS) volume

Throughput Optimized HDD EBS volumes are designed for frequently accessed, throughput-intensive workloads such as big data, data warehouses, and log processing. While they are cost-effective, they don't offer the consistent, low-latency performance required for this application.

Overall Explanation:

The application requires both durable storage and low-latency performance. Instance store volumes are not durable. Amazon ElastiCache is a caching service. Throughput Optimized HDD EBS volumes are not designed for low-latency performance. Provisioned IOPS SSD EBS volumes provide durable block storage with consistent, low-latency performance, making it the best choice.

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Question 16

Not Attempted

Storage



An online photo-sharing company stores its photos in an Amazon S3 bucket that exists in the us-west-1 Region. The company needs to store a copy of all new photos in the us-east-1 Region. Which solution will meet this requirement with the LEAST operational effort?

✓ Create a second S3 bucket in us-east-1. Use S3 Cross-Region Replication to copy photos from the existing S3 bucket to the second S3 bucket.

S3 Cross-Region Replication handles automatically copying new objects added to the source bucket to the destination bucket in a different region. It continuously replicates new photos without

needing to manually copy files or set up Lambda triggers. It provides automated replication that minimizes operational overhead.

- ⊗ Create a cross-origin resource sharing (CORS) configuration of the existing S3 bucket. Specify us-east-1 in the CORS rule's AllowedOrigin element.
CORS only enables cross-origin access; it does not copy objects.
- ⊗ Create a second S3 bucket in us-east-1 across multiple Availability Zones. Create an S3 Lifecycle rule to save photos into the second S3 bucket.
Using Lifecycle rules requires custom configuration and doesn't directly replicate the objects as they are created.
- ⊗ Create a second S3 bucket in us-east-1. Configure S3 event notifications on object creation and update events to invoke an AWS Lambda function to copy photos from the existing S3 bucket to the second S3 bucket.
Using Lambda functions requires custom code and logic to handle the copying, increasing operational overhead.

Overall Explanation:

S3 Cross-Region Replication is the most efficient solution as it automatically replicates new objects to a different region with the least operational overhead. Other options require manual intervention or custom code, increasing management complexity.

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Question 17

Not Attempted

Database



A company is developing a new web application for its subscribers. The application features a static single-page front end and a persistent database layer. The application is expected to handle millions of users during peak hours (4 hours in the morning) and a few thousand users for the remainder of the day. The data architects require the ability to rapidly evolve the database schema. Which combination of solutions will meet these requirements while providing the MOST scalability? (Choose two.)

- ✔ Deploy Amazon DynamoDB as the database solution. Provision on-demand capacity.
DynamoDB on-demand capacity provides the most scalability by instantly accommodating workload changes, scaling up or down as needed without pre-defined limits. This is ideal for

applications with fluctuating traffic patterns and the need to rapidly evolve the schema.

- ⊗ Deploy Amazon Aurora as the database solution. Choose the serverless DB engine mode.

Aurora Serverless is an on-demand, auto-scaling configuration for Amazon Aurora. While it provides scalability, it might not be the MOST scalable option compared to DynamoDB for schema evolution and handling highly variable workloads.

- ⊗ Deploy Amazon DynamoDB as the database solution. Ensure that DynamoDB auto scaling is enabled.

DynamoDB auto scaling modifies provisioned throughput settings over several minutes. While it scales, it's better suited for predictable traffic and gradual workload changes, not rapid scaling. It doesn't provide the MOST scalability compared to On-Demand capacity.

- ✓ Deploy the static content into an Amazon S3 bucket. Provision an Amazon CloudFront distribution with the S3 bucket as the origin.

Amazon S3 and CloudFront provide excellent scalability and performance for serving static content. CloudFront caches content globally, reducing latency for users and offloading traffic from the origin S3 bucket. This is the MOST scalable solution for hosting static content.

- ⊗ Deploy the web servers for static content across a fleet of Amazon EC2 instances in Auto Scaling groups. Configure the instances to periodically refresh the content from an Amazon Elastic File System (Amazon EFS) volume.

Using EC2 instances with Auto Scaling for static content is less scalable and more complex than using S3 and CloudFront. Managing a fleet of EC2 instances and handling content distribution and caching is less efficient.

Overall Explanation:

The best solution for handling the requirements of this application involves using DynamoDB with on-demand capacity for the database layer, as it allows for rapid schema evolution and instantaneous scalability based on the traffic load. For the static content, storing it in S3 and serving it via CloudFront provides the best scalability and global distribution, reducing load on the origin and delivering content quickly to users. Therefore options A and D are the correct answers.

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A company uses Amazon API Gateway to manage its REST APIs that third-party service providers access. The company must protect the REST APIs from SQL injection and cross-site scripting attacks. What is the MOST operationally efficient solution that meets these requirements?

☐ Configure AWS Shield.

AWS Shield primarily protects against DDoS attacks and doesn't directly address SQL injection or cross-site scripting.

☒ Configure AWS WAF.

AWS WAF is designed to protect web applications from common web exploits like SQL injection and cross-site scripting.

☐ Set up API Gateway with an Amazon CloudFront distribution. Configure AWS Shield in CloudFront.

While CloudFront and Shield can provide DDoS protection, WAF is more specific to the described threat and is a more direct solution.

☐ Set up API Gateway with an Amazon CloudFront distribution. Configure AWS WAF in CloudFront.

This option protects against SQL injection and XSS attacks, but the question asks for the *most operationally efficient* solution. Adding CloudFront introduces extra complexity and cost when the protection is needed at the API Gateway level.

Overall Explanation:

AWS WAF is the most operationally efficient solution because it directly addresses SQL injection and cross-site scripting attacks at the API Gateway level, without the added complexity of setting up a CloudFront distribution. Shield protects against DDOS attacks, while WAF is specifically designed to protect against SQL injection and XSS.

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Question 19

Not Attempted

Security



A company with 1,500 users manages on-premises resource access using Active Directory user groups. They want to grant users access to AWS resources without requiring them to maintain

separate credentials. A solutions architect needs to manage AWS resource access while preserving on-premises access. What is the MOST suitable solution?

- ⊗ Create an IAM user for each user in the company. Attach the appropriate policies to each user.

Creating individual IAM users for each employee is not scalable or manageable, especially with an existing Active Directory infrastructure. It also forces users to manage separate AWS credentials.

- ⊗ Use Amazon Cognito with an Active Directory user pool. Create roles with the appropriate policies attached.

While Cognito can integrate with Active Directory, it's primarily designed for managing user identities for mobile and web applications, not for granting access to AWS resources directly.

- ⊗ Define cross-account roles with the appropriate policies attached. Map the roles to the Active Directory groups.

Cross-account roles are used to grant access between AWS accounts, not to federate with on-premises Active Directory. While it could be a component of a more complex solution, it doesn't directly address the core requirement.

- ✔ Configure Security Assertion Markup Language (SAML) 2.0-based federation. Create roles with the appropriate policies attached. Map the roles to the Active Directory groups.

SAML 2.0 federation allows users to authenticate using their existing Active Directory credentials and then assume IAM roles to access AWS resources. This solution leverages the existing identity provider and avoids creating duplicate credentials.

Overall Explanation:

SAML 2.0-based federation is the BEST solution because it allows users to leverage their existing Active Directory credentials to access AWS resources without needing to create new ones. By mapping Active Directory groups to IAM roles, the architect can efficiently manage access based on existing on-premises group memberships. Cognito is more appropriate for web and mobile application authentication, and creating individual IAM users is not scalable or manageable. Cross-account roles address cross-account access, not on-premises federation.

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Question 20

Not Attempted

Networking



A company hosts a website behind multiple Application Load Balancers. The company possesses different distribution rights for its content across the globe. A solutions architect needs to ensure that users are served the appropriate content without violating these distribution rights. Which configuration should the solutions architect implement to meet these requirements?

☐ **Configure Amazon CloudFront with AWS WAF.**

While CloudFront is useful for content delivery and WAF can provide security, this option does not directly address the requirement of serving different content based on geographical distribution rights.

☐ **Configure Application Load Balancers with AWS WAF.**

ALBs and WAF can provide security, but they don't inherently offer geolocation-based content routing or restrictions.

☒ **Configure Amazon Route 53 with a geolocation policy.**

Amazon Route 53 with a geolocation policy allows routing traffic based on the user's location. This enables the company to direct users to specific Application Load Balancers serving content appropriate for their region, ensuring compliance with distribution rights.

☐ **Configure Amazon Route 53 with a geoproximity routing policy.**

Geoproximity routing primarily focuses on routing users to the 'nearest' resource, and doesn't allow explicitly restricting content access based on distribution rights for certain locations. It also allows for biasing traffic, which isn't the objective of the problem.

Overall Explanation:

The best solution is to configure Amazon Route 53 with a geolocation policy. This allows directing users to different Application Load Balancers based on their geographic location, ensuring they receive the correct content while respecting distribution rights. Geolocation routing lets you localize content and even restrict distribution based on licensing. Options A and B focus on security but don't address the content distribution requirements. Geoproximity routing (option D) doesn't easily enforce strict distribution rights by location.

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Question 21

Not Attempted

Storage



A company's data is growing beyond its on-premises capacity. The company wants to migrate this data to an Amazon S3 bucket. They need a solution that automatically validates the data's integrity after the transfer. Which solution meets these requirements?

- ☐ Order an AWS Snowball Edge device. Configure the Snowball Edge device to perform the online data transfer to an S3 bucket.

Snowball Edge is a physical device primarily for large-scale data transfers where network connectivity is limited. While it can transfer data, it doesn't inherently guarantee automatic data integrity validation after transfer.

- ☒ Deploy an AWS DataSync agent on premises. Configure the DataSync agent to perform the online data transfer to an S3 bucket.

AWS DataSync automatically checks data integrity during transfer. It can be configured to verify only the transferred data by calculating checksums of transferred files and metadata at the source location, meeting the requirement for automatic validation.

- ☐ Create an Amazon S3 File Gateway on premises. Configure the S3 File Gateway to perform the online data transfer to an S3 bucket.

S3 File Gateway primarily provides a local cache for frequently accessed data stored in S3 and handles file system protocols. It does not inherently provide automatic data integrity validation during the initial data transfer from on-premises to S3.

- ☐ Configure an accelerator in Amazon S3 Transfer Acceleration on premises. Configure the accelerator to perform the online data transfer to an S3 bucket.

S3 Transfer Acceleration optimizes data transfer speeds to S3 over long distances by utilizing CloudFront's edge locations. However, it focuses on speed and doesn't guarantee automatic data integrity validation after the transfer is complete.

Overall Explanation:

AWS DataSync is designed to securely and efficiently transfer data between on-premises storage and AWS services. A key feature is its automatic data integrity validation during transfer, calculating checksums to ensure data consistency. The other options either don't focus on data integrity validation (Snowball Edge, S3 Transfer Acceleration) or primarily serve a different purpose (S3 File Gateway).

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Question 22

Not Attempted

Networking



A company wants to migrate two DNS servers to AWS. The servers host a total of approximately 200 zones and receive 1 million requests each day on average. The company wants to maximize availability while minimizing the operational overhead that is related to the management of the two servers. Which solution should a solutions architect recommend to meet these requirements?

- ☒ Create 200 new hosted zones in the Amazon Route 53 console. Import zone files.
This option leverages Route 53, a highly available and scalable DNS service, minimizing operational overhead by offloading DNS server management. Importing zone files simplifies the migration process.
- ☐ Launch a single large Amazon EC2 instance. Import zone files. Configure Amazon CloudWatch alarms and notifications to alert the company about any downtime.
This option does not maximize availability, as a single EC2 instance represents a single point of failure. It also increases operational overhead with instance management and monitoring.
- ☐ Migrate the servers to AWS by using AWS Server Migration Service (AWS SMS). Configure Amazon CloudWatch alarms and notifications to alert the company about any downtime.
This option does not inherently maximize availability. It also involves managing and maintaining EC2 instances, increasing operational overhead.
- ☐ Launch an Amazon EC2 instance in an Auto Scaling group across two Availability Zones. Import zone files. Set the desired capacity to 1 and the maximum capacity to 3 for the Auto Scaling group. Configure scaling alarms to scale based on CPU utilization.
While this option increases availability, it still involves managing EC2 instances and scaling policies, leading to higher operational overhead than using a managed service like Route 53. Also, scaling based on CPU utilization is not optimal for DNS.

Overall Explanation:

The best solution is to use Amazon Route 53. Route 53 is a highly available and scalable DNS service. It minimizes operational overhead by offloading DNS server management.

The zone files can be directly imported into Route 53. Options B, C, and D require more management and are not as highly available.

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Question 23

Not Attempted

Storage

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A global company operates its applications across multiple AWS accounts within an AWS Organization. These applications utilize multipart uploads to store data in Amazon S3 buckets spanning various AWS Regions. The company needs a solution to report on incomplete multipart uploads to optimize costs and ensure compliance. The solution should minimize operational overhead.

- ❌ **Configure AWS Config with a rule to report the incomplete multipart upload object count.**

AWS Config rules can be complex to set up and maintain for cross-account reporting and may require custom logic for multipart upload analysis, resulting in higher operational overhead.
- ❌ **Create a service control policy (SCP) to report the incomplete multipart upload object count.**

SCPs primarily focus on permissions and governance; they cannot directly report on object-level details like incomplete multipart uploads. SCPs are not designed for reporting or data analysis.
- ✅ **Configure S3 Storage Lens to report the incomplete multipart upload object count.**

S3 Storage Lens provides organization-wide visibility into object storage, including built-in metrics for incomplete multipart uploads, minimizing operational overhead. It offers cost-efficiency metrics specifically for analyzing incomplete multipart uploads.
- ❌ **Create an S3 Multi-Region Access Point to report the incomplete multipart upload object count.**

S3 Multi-Region Access Points are designed to provide a single endpoint for accessing data across multiple S3 buckets in different regions, focusing on availability and performance. They do not inherently offer reporting on incomplete multipart uploads.

Overall Explanation:

S3 Storage Lens is the most suitable option. It is a cloud storage analytics feature that supports AWS Organizations, providing organization-wide visibility into object storage. It

has built-in metrics and trend lines, and it automatically provides actionable recommendations. Specifically, it offers Cost Efficiency metrics, including 'Incomplete Multipart Upload Object Count,' to directly address the requirement with minimal operational overhead. Therefore, it is the right answer.

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Question 24

Not Attempted

Database



A company runs a production database on Amazon RDS for MySQL. The company wants to upgrade the database version for security compliance reasons. Because the database contains critical data, the company wants a quick solution to upgrade and test functionality without losing any data and with the LEAST operational overhead.

- ☐ Create an RDS manual snapshot. Upgrade to the new version of Amazon RDS for MySQL.
This option involves manual steps and might not be the quickest or least operationally intensive approach, especially for testing.
- ☐ Use native backup and restore. Restore the data to the upgraded new version of Amazon RDS for MySQL.
This involves manual backup and restore, which can be time-consuming and adds operational overhead.
- ☐ Use AWS Database Migration Service (AWS DMS) to replicate the data to the upgraded new version of Amazon RDS for MySQL.
DMS is suitable for migrations but adds more complexity and overhead than a solution specifically designed for in-place upgrades.
- ☒ Use Amazon RDS Blue/Green Deployments to deploy and test production changes.
Blue/Green deployments provide a managed solution for upgrading and testing with minimal downtime and operational overhead. It allows for thorough testing in a separate environment before promoting to production.

Overall Explanation:

Amazon RDS Blue/Green Deployments offer a streamlined, managed approach to upgrade database versions. It minimizes downtime, allows thorough testing in a separate

environment, and reduces operational overhead compared to manual snapshot/restore or DMS-based migration. A blue/green deployment copies the production database environment to a separate, synchronized staging environment. Changes can be made to the database in the staging environment without affecting the production environment. When ready, the staging environment can be promoted to be the new production database environment, with downtime typically under one minute.

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Question 25

Not Attempted

Storage



A company hosts its infrastructure on AWS and supports 700,000 users for its document management application. They plan to introduce a new feature to convert large .pdf files (average size 5 MB) into .jpg image files. Both the original .pdf files and the converted .jpg files need to be stored. A solutions architect needs to design a cost-effective and scalable solution to handle rapidly growing demand. Which solution BEST meets these requirements?

- ☒ Save the .pdf files to Amazon S3. Configure an S3 PUT event to invoke an AWS Lambda function to convert the files to .jpg format and store them back in Amazon S3.

This is the most cost-effective solution. S3 offers scalable and cost-effective storage, and Lambda provides serverless compute that scales automatically and charges only for the compute time used. The S3 PUT event trigger removes the need to manage infrastructure for the conversion process.

- ☐ Save the .pdf files to Amazon DynamoDB. Use the DynamoDB Streams feature to invoke an AWS Lambda function to convert the files to .jpg format and store them back in DynamoDB.

DynamoDB is not suitable for storing large files like .pdf documents and would be very expensive.

- ☐ Upload the .pdf files to an AWS Elastic Beanstalk application with Amazon EC2 instances, Amazon Elastic Block Store (Amazon EBS) storage, and an Auto Scaling group. Use a program in the EC2 instances to convert the files to .jpg format. Save the .pdf files and the .jpg files in the EBS store.

Elastic Beanstalk with EC2 and EBS is a viable option but is more costly and involves managing the underlying infrastructure. EBS storage is also not as cost-effective as S3 for this use case.

- ☐ Upload the .pdf files to an AWS Elastic Beanstalk application with Amazon EC2 instances, Amazon Elastic File System (Amazon EFS) storage, and an Auto Scaling

group. Use a program in the EC2 instances to convert the file to .jpg format. Save the .pdf files and the .jpg files in the EFS store.

Elastic Beanstalk with EC2 and EFS is less cost-effective. EFS is shared file storage, and scaling and performance may not be optimal, especially with increasing demand and file sizes. Also requires managing the underlying infrastructure.

Overall Explanation:

The most cost-effective solution is to leverage Amazon S3 for storing the files and AWS Lambda for the conversion process. S3 provides scalable and inexpensive storage, and Lambda offers serverless compute that scales automatically. The S3 PUT event triggering Lambda eliminates the need for managing infrastructure, making Option A the best choice.

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Question 26

Not Attempted

Serverless



A solutions architect is designing a data processing job that must run once daily and can take up to 2 hours to complete. The job needs to restart from the beginning if interrupted. What is the MOST cost-effective way to address this requirement?

- ☐ Create a script that runs locally on an Amazon EC2 Reserved Instance that is triggered by a cron job.
Using EC2 Reserved Instances can be cost-effective for long-term, consistent workloads. However, running a script directly on an EC2 instance might not be the most resilient or scalable solution, and Reserved Instances are charged regardless of usage.
- ☐ Create an AWS Lambda function triggered by an Amazon EventBridge scheduled event.
AWS Lambda functions have a maximum execution time limit (currently 15 minutes). Therefore, a Lambda function is not suitable for a job that can take up to 2 hours to complete.
- ☒ Use an Amazon Elastic Container Service (Amazon ECS) Fargate task triggered by an Amazon EventBridge scheduled event.

Amazon ECS Fargate allows you to run containers without managing the underlying infrastructure. It bills you for the resources consumed during the task execution, making it a cost-effective option for tasks with variable durations. It's suitable for a job that takes up to 2 hours and can be restarted from the beginning if interrupted.

- ☐ Use an Amazon Elastic Container Service (Amazon ECS) task running on Amazon EC2 triggered by an Amazon EventBridge scheduled event.

Using Amazon ECS with EC2 instances requires you to manage the EC2 instances, including patching and scaling. Also, EC2 instances will be billed regardless of the task duration. This is less cost-effective than Fargate, which charges only for the duration of the task.

Overall Explanation:

Amazon ECS Fargate provides a cost-effective way to run containerized applications without managing the underlying infrastructure. It's billed per task execution time, making it suitable for jobs with variable durations that need to be restarted if interrupted.

Lambda's time limit makes it unsuitable, and using EC2 directly or with ECS is less cost-effective than Fargate. Therefore, ECS Fargate is the best choice.

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Question 27

Not Attempted

Database



A social media company wants to store its database of user profiles, relationships, and interactions in the AWS Cloud. The company needs an application to monitor any changes in the database. The application needs to analyze the relationships between the data entities and to provide recommendations to users. Which solution will meet these requirements with the LEAST operational overhead?

- ☐ Use Amazon Neptune to store the information. Use Amazon Kinesis Data Streams to process changes in the database.

Kinesis Data Streams can process changes, but Neptune Streams offers a more integrated and less operationally complex solution for graph data changes.

- ☒ Use Amazon Neptune to store the information. Use Neptune Streams to process changes in the database.

Amazon Neptune is designed for graph data, making it suitable for storing relationships. Neptune Streams provides a managed way to track changes in the graph database with minimal operational overhead.

- ☐ Use Amazon Quantum Ledger Database (Amazon QLDB) to store the information. Use Amazon Kinesis Data Streams to process changes in the database.

Amazon QLDB is designed for maintaining an immutable and auditable transaction history, not for analyzing relationships in graph data. Kinesis Streams alone doesn't provide the necessary graph analysis capabilities.

- ⊗ Use Amazon Quantum Ledger Database (Amazon QLDB) to store the information. Use Neptune Streams to process changes in the database.

Amazon QLDB is not designed for graph data. Neptune Streams can only be used to track changes in Neptune.

Overall Explanation:

Amazon Neptune is a graph database suitable for managing relationships in social media data. Neptune Streams is a managed service that captures a sequence of change-log entries that record every change made to your graph data. Using Neptune and Neptune Streams minimizes the operational overhead compared to other solutions.

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Question 28

Not Attempted

Storage



A company is developing a new application that needs to store and process a large and growing amount of data. The data will be analyzed hourly and modified by several Amazon EC2 Linux instances distributed across multiple Availability Zones. The required storage capacity is expected to increase significantly over the next 6 months. Which storage solution best fulfills these requirements?

- ⊗ Store the data in Amazon S3 Glacier. Update the S3 Glacier vault policy to allow access to the application instances.

S3 Glacier is designed for long-term archival storage with infrequent access. It is not suitable for hourly data analysis and modification.

- ⊗ Store the data in an Amazon Elastic Block Store (Amazon EBS) volume. Mount the EBS volume on the application instances.

EBS volumes can only be attached to a single EC2 instance at a time and are tied to a specific Availability Zone, making them unsuitable for multi-AZ access by multiple instances.

- ✔ Store the data in an Amazon Elastic File System (Amazon EFS) file system. Mount the file system on the application instances.

EFS provides a shared file system that can be mounted by multiple EC2 instances concurrently, even across different Availability Zones. It also automatically scales storage capacity as needed, fulfilling the stated requirements.

- ⊗ Store the data in an Amazon Elastic Block Store (Amazon EBS) Provisioned IOPS volume shared between the application instances.

EBS volumes cannot be shared between multiple EC2 instances across different Availability Zones. While Provisioned IOPS improve performance, this solution doesn't meet the AZ requirement.

Overall Explanation:

Amazon EFS is the most suitable solution because it allows multiple EC2 instances to access and modify data concurrently across multiple Availability Zones. EFS automatically scales storage capacity as needed, and unlike S3 Glacier, it's designed for frequently accessed data. EBS volumes cannot be shared across Availability Zones.

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Question 29

Not Attempted

Database



A company's application, which relies on an Amazon RDS for PostgreSQL Multi-AZ DB instance, is facing performance bottlenecks due to increased traffic and heavy database query loads.

What should a solutions architect recommend to alleviate the performance issues?

- ⊗ Serve read traffic from the Multi-AZ standby replica.

A Multi-AZ DB instance provides high availability through a primary DB instance and a standby in a different Availability Zone. However, the standby instance does not support connections for read workloads. Therefore, directing read traffic to the standby is not a viable solution.

- ⊗ Configure the DB instance to use Transfer Acceleration.

Transfer Acceleration is designed to expedite traffic to and from Amazon S3. It is not applicable or helpful in addressing database performance issues related to query load on an RDS instance.

- ✓ Create a read replica from the source DB instance. Serve read traffic from the read replica.

Creating a read replica from the primary DB instance allows you to offload read traffic to the replica. This distributes the database load and improves performance by reducing the load on the primary instance.

- ⊗ Use Amazon Kinesis Data Firehose between the application and Amazon RDS to increase the concurrency of database requests.

Kinesis Data Firehose is a service for loading streaming data into data stores and analytics tools. While it can handle high volumes of data, it doesn't directly address the performance issues caused by heavy read query loads on the RDS instance. It also won't solve DB performance issues.

Overall Explanation:

The optimal solution to improve the application's performance is to create a read replica from the source DB instance and direct read traffic to the read replica. This approach offloads the read workload from the primary DB instance, enhancing overall performance. Multi-AZ standby replica cannot serve traffic. Transfer Acceleration is for S3, and Kinesis Firehose is not applicable to this scenario.

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Question 30

Not Attempted

Storage



A company gathers 10 GB of telemetry data daily from various machines and stores it in an Amazon S3 bucket within a source data account. Several consulting agencies have been contracted to analyze this data and require read access. The company wants to share the data securely and efficiently from the source data account. Which solution best meets these requirements?

- ⊗ Configure S3 global tables to replicate data for each agency.

S3 Global Tables are designed for low-latency access to data across geographically dispersed locations, not for sharing data with external agencies for analysis. This option is not suitable and practically doesn't exist.

- ⊗ Make the S3 bucket public for a limited time and inform the agencies.

Making the S3 bucket public exposes the data to anyone on the internet, creating a significant security risk. This is not a recommended approach.

- ✔ Configure cross-account access for the S3 bucket to the AWS accounts that the agencies own.

Cross-account access allows the company to grant specific permissions to the agencies' AWS accounts, enabling their analysts to access the data in the S3 bucket securely and efficiently. This is the most secure and operationally efficient solution.

- ⊗ Set up an IAM user for each analyst in the source data account. Grant each user access to the S3 bucket.

Creating individual IAM users for external analysts within the company's account introduces management overhead and potential security risks. Managing numerous external user accounts is complex and less secure than cross-account access.

Overall Explanation:

Configuring cross-account access (Option C) is the best solution. It enables secure access to the S3 bucket for the consulting agencies' AWS accounts without compromising the company's security posture or creating unnecessary management overhead. Options A and B are unsuitable and insecure, while Option D is less secure and operationally inefficient compared to Option C. The provided AWS documentation link exemplifies how to implement cross-account S3 bucket access, ensuring each agency only has access to what they need.

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Question 31

Not Attempted

Storage



A company uses Amazon FSx for NetApp ONTAP in its primary AWS Region for CIFS and NFS file shares. Applications that run on Amazon EC2 instances access the file shares. The company needs a storage disaster recovery (DR) solution in a secondary Region. The data that is replicated in the secondary Region needs to be accessed by using the same protocols as the primary Region. Which solution will meet these requirements with the LEAST operational overhead?

- ⊗ Create an AWS Lambda function to copy the data to an Amazon S3 bucket. Replicate the S3 bucket to the secondary Region.

This solution is not compatible because it does not allow access with CIFS (SMB) or NFS protocols in the secondary region. S3 is an object store and requires significant application changes to work with it.

- ⊗ Create a backup of the FSx for ONTAP volumes by using AWS Backup. Copy the volumes to the secondary Region. Create a new FSx for ONTAP instance from the backup.

This involves a one-time copy and does not provide continuous replication for disaster recovery. The restore process also adds operational overhead.

- ✔ Create an FSx for ONTAP instance in the secondary Region. Use NetApp SnapMirror to replicate data from the primary Region to the secondary Region.

This is the recommended solution. NetApp SnapMirror provides a managed and efficient way to replicate data between FSx for ONTAP file systems, including cross-region replication, while maintaining protocol compatibility (CIFS/NFS).

- ✘ Create an Amazon Elastic File System (Amazon EFS) volume. Migrate the current data to the volume. Replicate the volume to the secondary Region.

EFS does not offer SMB (CIFS) protocol support. Therefore, it cannot meet the requirement of using the same protocols as the primary region.

Overall Explanation:

NetApp SnapMirror, integrated with Amazon FSx for NetApp ONTAP, provides the least operational overhead for replicating data to a secondary region for DR purposes while maintaining CIFS/NFS protocol compatibility. It offers a managed solution for continuous replication with low RPO and RTO. Options A, B, and D do not meet all the requirements. Option A changes the storage target, Option B requires restoring from backup, and Option D does not support CIFS.

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Question 32

Not Attempted

Serverless



A development team is creating an event-based application that uses AWS Lambda functions. Events are generated when files are added to an Amazon S3 bucket. The development team currently has Amazon Simple Notification Service (Amazon SNS) configured as the event target from Amazon S3. To ensure scalable processing of events from Amazon S3, what should a solutions architect implement?

- ✘ Create an SNS subscription that processes the event in Amazon Elastic Container Service (Amazon ECS) before the event runs in Lambda.

Using ECS solely for pre-processing SNS events before Lambda introduces unnecessary complexity and overhead.

- ☐ Create an SNS subscription that processes the event in Amazon Elastic Kubernetes Service (Amazon EKS) before the event runs in Lambda.
Similar to ECS, using EKS for pre-processing adds complexity without significantly improving scalability for this event-driven scenario.
- ☒ Create an SNS subscription that sends the event to Amazon Simple Queue Service (Amazon SQS). Configure the SQS queue to trigger a Lambda function.
This is the most scalable solution. SNS pushes messages to SQS, which decouples the event source from the Lambda function. SQS can handle a large volume of messages and Lambda can scale automatically when triggered by SQS.
- ☐ Create an SNS subscription that sends the event to AWS Server Migration Service (AWS SMS). Configure the Lambda function to poll from the SMS event.
AWS SMS is used for migrating servers, not for general event processing. It is not the appropriate service for this use case.

Overall Explanation:

The most scalable approach is to leverage Amazon SQS as an intermediary between SNS and Lambda. SNS publishes notifications to SQS, and SQS acts as a buffer and decouples the event source from the Lambda function. This allows Lambda to scale efficiently based on the number of messages in the queue, providing better load distribution and resilience compared to directly invoking Lambda from SNS. Amazon SQS is designed for event-driven and scalable message processing.

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Question 33

Not Attempted

[Serverless](#)

A solutions architect is designing a new service behind Amazon API Gateway. The request patterns for the service will be unpredictable and can change suddenly from 0 requests to over 500 per second. The total size of the data that needs to be persisted in a backend database is currently less than 1 GB with unpredictable future growth. Data can be queried using simple key-value requests. Which combination of AWS services would meet these requirements? (Choose two.)

- ☐ AWS Fargate

AWS Fargate, while scalable, typically involves ECS or EKS which adds unnecessary complexity for a service needing to scale from 0 to 500 requests per second. Lambda provides a more suitable 'scale to zero' capability.

✓ AWS Lambda

AWS Lambda is ideal for handling unpredictable and sudden changes in request patterns as it can scale from 0 to many requests per second automatically.

✓ Amazon DynamoDB

Amazon DynamoDB is a NoSQL database that scales automatically and is well-suited for key-value data storage with unpredictable growth and simple query requirements.

✗ Amazon EC2 Auto Scaling

Amazon EC2 Auto Scaling might not be as responsive as Lambda for scaling from zero to hundreds of requests per second, making it less suitable for highly variable workloads.

✗ MySQL-compatible Amazon Aurora

MySQL-compatible Amazon Aurora is a relational database and is not ideal for simple key-value data storage. DynamoDB is a better choice.

Overall Explanation:

AWS Lambda is the best choice for handling unpredictable request patterns due to its ability to scale quickly and efficiently. Amazon DynamoDB is the optimal database solution for key-value data with unpredictable growth, providing scalability and simple query capabilities.

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Question 34

Not Attempted

Storage



A global research company wants to collect, store, and share research data with its employees worldwide. The company intends to store the data securely in an Amazon S3 bucket and process it within the AWS Cloud. Employees require secure access to both upload and download data. The company is looking for a solution that minimizes operational overhead. Which solution best meets these requirements?

- ⊗ Use an AWS Lambda function to create S3 presigned URLs. Instruct employees to use these URLs for data access.

Using presigned URLs requires frequent regeneration and distribution, leading to increased operational overhead and management complexity. Presigned URLs are temporary, requiring regular renewal and employee instruction.

- ⊗ Create an IAM user for each employee. Create specific IAM policies for each employee to grant S3 access. Instruct employees to use the AWS Management Console.

Creating and managing individual IAM users for each employee is operationally intensive and doesn't scale efficiently. Manually using the AWS Management Console for file transfer can be complex and less user-friendly.

- ✓ Create an S3 File Gateway. Configure a share for uploading and a share for downloading. Allow employees to mount these shares on their local computers for seamless access to the S3 bucket.

S3 File Gateway allows users to mount the S3 bucket as a network drive, simplifying file uploads and downloads. It is a simple access for uploading/downloading with low operational overhead.

- ⊗ Configure AWS Transfer Family SFTP endpoints. Select custom identity provider options. Use AWS Secrets Manager to manage user credentials. Instruct employees to use an SFTP client to access Transfer Family.

While secure, using AWS Transfer Family with custom identity providers and Secrets Manager is more complex to set up and maintain than using S3 File Gateway. Secrets Manager is not for managing user credentials.

Overall Explanation:

S3 File Gateway provides a relatively simple and operationally efficient way for employees to access data stored in S3. Employees can easily mount shares on their computers for seamless uploading and downloading, minimizing operational overhead compared to managing individual IAM users or presigned URLs. AWS Transfer Family is a good choice but can be more complex than needed. Lambda functions for presigned URLs involve manual generation and management. Creating individual IAM users adds significant operational overhead.

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A company is building a new furniture inventory application. The company has deployed the application on a fleet of Amazon EC2 instances across multiple Availability Zones. The EC2 instances run behind an Application Load Balancer (ALB) in their VPC. A solutions architect has observed that incoming traffic seems to favor one EC2 instance, resulting in latency for some requests. What should the solutions architect do to resolve this issue?

☒ **Disable session affinity (sticky sessions) on the ALB**

Disabling session affinity will cause the ALB to distribute requests across all EC2 instances in the target group, balancing the load and reducing latency. When session affinity is enabled, the ALB directs requests from the same client to the same instance, which can lead to uneven load distribution.

☐ **Replace the ALB with a Network Load Balancer**

Replacing the ALB with a Network Load Balancer won't directly address the issue of traffic favoring one instance. NLBs operate at Layer 4 and do not have session affinity features that cause this behavior. It may introduce complexities for HTTP/HTTPS traffic.

☐ **Increase the number of EC2 instances in each Availability Zone**

While increasing the number of EC2 instances can help overall capacity and availability, it doesn't directly solve the problem of traffic being unevenly distributed among existing instances. The root cause is likely session affinity.

☐ **Adjust the frequency of the health checks on the ALB's target group**

Adjusting health check frequency will help ensure unhealthy instances are taken out of rotation, but it does not address the problem of traffic being favored on one instance while healthy instances are available.

Overall Explanation:

The issue of traffic favoring one EC2 instance is likely due to session affinity (sticky sessions) being enabled on the Application Load Balancer (ALB). Disabling session affinity allows the ALB to distribute traffic evenly across all healthy instances in the target group, resolving the latency issue. Options B, C, and D do not directly address the root cause of the problem.

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A company with over 5 TB of file data on on-premises Windows file servers is migrating Windows workloads to AWS. The company requires low-latency access to both AWS and on-premises file storage during the migration. The solution must minimize operational overhead and avoid significant changes to existing file access patterns. Connectivity to AWS is established through an AWS Site-to-Site VPN connection. What should a solutions architect do to meet these requirements?

- ⊗ Deploy and configure Amazon FSx for Windows File Server on AWS. Move the on-premises file data to FSx for Windows File Server. Reconfigure the workloads to use FSx for Windows File Server on AWS.

This option may not provide low latency access for on-premises users, as they would be accessing the data over the VPN connection. Also, it requires moving all data to AWS.

- ⊗ Deploy and configure an Amazon S3 File Gateway on premises. Move the on-premises file data to the S3 File Gateway. Reconfigure the on-premises workloads and the cloud workloads to use the S3 File Gateway.

S3 File Gateway doesn't provide native Windows file server compatibility, requiring significant changes to existing file access patterns.

- ⊗ Deploy and configure an Amazon S3 File Gateway on premises. Move the on-premises file data to Amazon S3. Reconfigure the workloads to use either Amazon S3 directly or the S3 File Gateway depending on each workload's location.

Similar to option B, this solution requires significant changes to existing file access patterns because the application would need to access object storage rather than a file share.

- ✔ Deploy and configure Amazon FSx for Windows File Server on AWS. Deploy and configure an Amazon FSx File Gateway on premises. Move the on-premises file data to the FSx File Gateway. Configure the cloud workloads to use FSx for Windows File Server on AWS. Configure the on-premises workloads to use the FSx File Gateway.

This is the correct answer because FSx File Gateway provides low-latency access to the in-cloud FSx for Windows File Server for on-premise workloads, minimizing operational overhead and preserving existing file access patterns. The Gateway caches frequently accessed files, providing faster performance than accessing the file share over a VPN.

Overall Explanation:

The best solution is to deploy Amazon FSx for Windows File Server on AWS and an Amazon FSx File Gateway on-premises. The FSx File Gateway will provide low-latency access to the data in the AWS Cloud, while the cloud workloads can access the FSx file

share directly in AWS. This approach minimizes changes to existing file access patterns and reduces operational overhead by leveraging managed services.

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Question 37

Not Attempted

Security



A company has an application workflow that uses an AWS Lambda function to download and decrypt files from Amazon S3. These files are encrypted using AWS Key Management Service (AWS KMS) keys. A solutions architect needs to design a solution that will ensure the required permissions are set correctly for the Lambda function to decrypt the KMS encrypted files. Which combination of actions accomplish this? (Choose two.)

- ☐ Attach the kms:decrypt permission to the Lambda function's resource policy.
The Lambda function's resource policy controls who can invoke the function, not what the function can do. This option is incorrect.
- ☒ Grant the decrypt permission for the Lambda IAM role in the KMS key's policy.
This is correct. The KMS key policy needs to allow the Lambda function's IAM role to decrypt using the key.
- ☐ Grant the decrypt permission for the Lambda resource policy in the KMS key's policy.
The Lambda function's resource policy is not the entity that needs permission to decrypt. The IAM role that the Lambda function assumes needs the permission.
- ☐ Create a new IAM policy with the kms:decrypt permission and attach the policy to the Lambda function.
While creating an IAM policy is part of the solution, you attach it to the IAM *role*, not directly to the Lambda function.
- ☒ Create a new IAM role with the kms:decrypt permission and attach the execution role to the Lambda function.
It's more appropriate to modify existing IAM role attached to lambda, rather than creating a new one. It is also necessary to give lambda IAM role permissions on KMS key to decrypt

Overall Explanation:

The correct approach involves two key steps. First, grant the Lambda function's IAM role permission to decrypt using the KMS key by modifying the KMS key policy. Second, the Lambda function needs the `*kms:Decrypt*` permission, and that permission is attached to the execution IAM role of the Lambda function through IAM policy. The resource policy doesn't control what the function can do.

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Question 38

Not Attempted

Monitoring



A company needs to monitor its AWS costs for monthly financial reviews. The cloud operations team is designing an architecture within the AWS Organizations management account to query AWS Cost and Usage Reports (CUR) for all member accounts. The team is required to run this query once a month and provide a detailed bill analysis. Which solution offers the MOST scalable and cost-effective approach to meet these requirements?

- ☐ Enable Cost and Usage Reports in the management account. Deliver reports to Amazon Kinesis. Use Amazon EMR for analysis.

Using Kinesis and EMR is overkill for a monthly query. Neither Kinesis nor EMR is cost-effective for infrequent processing of cost and usage data. EMR is more suitable for large-scale data processing.

- ☒ Enable Cost and Usage Reports in the management account. Deliver the reports to Amazon S3. Use Amazon Athena for analysis.

This is the most scalable and cost-effective solution. Cost and Usage Reports are optimized for S3 delivery, and Athena directly integrates with S3, allowing for efficient querying of the data on a monthly basis.

- ☐ Enable Cost and Usage Reports for member accounts. Deliver the reports to Amazon S3. Use Amazon Redshift for analysis.

Delivering reports for each member account creates redundant data. This means extra time, effort, and cost, as well as additional accounts to manage. Redshift is more appropriate for large, complex data warehousing workloads, not infrequent monthly queries.

- ☐ Enable Cost and Usage Reports for member accounts. Deliver the reports to Amazon Kinesis. Use Amazon QuickSight for analysis.

Similar to option A and C, enabling CUR on each member account will create redundant data. This means extra time, extra work to do manually, extra money to cost, and extra account to manage. While QuickSight is a good BI tool, Kinesis is not cost effective for infrequent cost data processing.

Overall Explanation:

The most scalable and cost-effective solution involves enabling Cost and Usage Reports in the management account and delivering them to Amazon S3. Amazon Athena provides a serverless query engine that can directly analyze the data in S3, providing a cost-effective and scalable way to perform monthly cost analysis. This is because Cost and Usage Reports are already optimized for S3 delivery, and Athena directly integrates with S3. Using Kinesis and EMR, or Redshift, adds unnecessary complexity and cost for a monthly analysis. Enabling CUR on each member account also increases management complexity and data redundancy.

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Question 39

Not Attempted

Networking



A company is hosting a real-time multiplayer game on Amazon EC2 instances within an Auto Scaling group. The application uses UDP for communication to minimize latency. They need a solution to distribute the game traffic across the available EC2 instances and ensure high availability as the player base scales dynamically. Which solution best meets these requirements?



Attach a Network Load Balancer to the Auto Scaling group.

Network Load Balancers are designed to handle UDP traffic efficiently and can dynamically scale to accommodate changing traffic patterns. They operate at Layer 4 of the OSI model, providing low-latency forwarding suitable for real-time applications.



Attach an Application Load Balancer to the Auto Scaling group.

Application Load Balancers operate at Layer 7 and are designed for HTTP/HTTPS traffic. They do not support UDP, making them unsuitable for this scenario.



Deploy an Amazon Route 53 record set with a weighted policy to route traffic appropriately.

Route 53 is a DNS service and primarily focuses on domain name resolution. While weighted policies can distribute traffic, they are not ideal for real-time load balancing based on application health and capacity.



Deploy a NAT instance that is configured with port forwarding to the EC2 instances in the Auto Scaling group.

NAT instances are used for allowing instances in a private subnet to access the internet. They are not designed for load balancing and would not provide the scalability and high availability required for the gaming application.

Overall Explanation:

The correct answer is to attach a Network Load Balancer (NLB) to the Auto Scaling group. NLBs are specifically designed for handling UDP traffic and provide the necessary scalability and low-latency forwarding required for real-time gaming applications. They seamlessly integrate with Auto Scaling groups to dynamically adjust the distribution of traffic as instances are added or removed.

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Question 40

Not Attempted

Database



A company operates multiple websites on AWS, each generating tens of gigabytes of web traffic logs daily. A solutions architect needs to design a scalable and cost-effective solution that allows the company's developers to analyze traffic patterns across all websites. The analysis will be performed on demand, approximately once a week, over several months and must support standard SQL queries.



Store the logs in Amazon S3. Use Amazon Athena for analysis.

Storing logs in S3 and using Athena provides a scalable, serverless, and cost-effective solution for analyzing data using standard SQL. Athena directly queries data in S3, eliminating the need to load data into a database.



Store the logs in Amazon RDS. Use a database client for analysis.

Amazon RDS is not designed for large volumes of log data and frequent analytical queries. It is more suitable for transactional workloads and might not be as scalable or cost-effective as an S3-based solution.



Store the logs in Amazon OpenSearch Service. Use OpenSearch Service for analysis.

While OpenSearch Service can analyze logs, it might be more expensive than using Amazon S3 and Athena for infrequent analysis. Also, while OpenSearch does support SQL queries, it is primarily a search and analytics engine and is not optimized for ad-hoc SQL analysis in the same way as Athena. Furthermore, S3 storage is cheaper than OpenSearch storage.

- ⊗ Store the logs in an Amazon EMR cluster. Use a supported open-source framework for SQL-based analysis.

Using Amazon EMR for storing logs is not ideal, as EMR is more suited for processing data stored elsewhere. While EMR can process data with SQL-like frameworks, it's overkill and more expensive for this infrequent, on-demand analysis requirement.

Overall Explanation:

The most cost-effective solution is to store the logs in Amazon S3 and use Amazon Athena for analysis. Athena allows querying data directly in S3 using standard SQL, is serverless, and charges only for the queries run. RDS is not designed for this kind of workload. OpenSearch is more expensive, and EMR is unnecessary for on-demand analysis.

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Question 41

Not Attempted

Networking



An international company operates in multiple countries and has a subdomain for each country, such as `example.com`, `country1.example.com`, and `country2.example.com`. Workloads are behind an Application Load Balancer (ALB). The company needs to ensure that website data is encrypted in transit. What combination of actions will accomplish this?

- ✔ Use the AWS Certificate Manager (ACM) console to request a public certificate for the apex domain `example.com` and a wildcard certificate for `\*.example.com`.
- A public certificate is required for external communication and the wildcard certificate covers all subdomains.
- ⊗ Use the AWS Certificate Manager (ACM) console to request a private certificate for the apex domain `example.com` and a wildcard certificate for `\*.example.com`.
- Private certificates are for internal use and not trusted by external clients. This is incorrect.
- ⊗ Use the AWS Certificate Manager (ACM) console to request a public and private certificate for the apex domain `example.com`.
- Requesting both public and private certificates for only the apex domain does not provide the required wildcard coverage for subdomains.
- ⊗ Validate domain ownership by email address. Switch to DNS validation by adding the required DNS records to the DNS provider.

Validating domain ownership is crucial for ACM to issue certificates. While email validation is an option, DNS validation is preferable.

- ✔ Validate domain ownership for the domain by adding the required DNS records to the DNS provider.

DNS validation provides automated renewal and is a best practice for ACM certificate management.

Overall Explanation:

To encrypt website data in transit for multiple subdomains behind an ALB, a public certificate with a wildcard domain is required. ACM must validate domain ownership to issue the certificate. The best method to validate is through DNS, by adding the required DNS records to the DNS provider. Requesting a public certificate with a wildcard and validating through DNS will fulfill the requirements.

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Question 42

Not Attempted

Security



A company is required to use cryptographic keys in its on-premises key manager. The key manager is outside of the AWS Cloud due to regulatory and compliance requirements. The company wants to manage encryption and decryption using cryptographic keys retained outside the AWS Cloud and support a variety of external key managers from different vendors. Which solution will meet these requirements with the LEAST operational overhead?

- ✘ Use AWS CloudHSM key store backed by a CloudHSM cluster.
AWS CloudHSM resources reside within the AWS Cloud, failing the requirement to maintain keys outside of AWS.
- ✔ Use an AWS Key Management Service (AWS KMS) external key store backed by an external key manager.
An AWS KMS external key store allows you to use an external key manager outside of AWS to perform cryptographic operations. This solution has less operational overhead as AWS KMS integrates with the external key manager.
- ✘ Use the default AWS Key Management Service (AWS KMS) managed key store.
The default AWS KMS key store manages keys within AWS, failing the requirement to maintain keys outside of AWS.

- ☐ Use a custom key store backed by an AWS CloudHSM cluster.

A custom key store backed by CloudHSM stores keys within AWS CloudHSM, failing the requirement to maintain keys outside of AWS.

Overall Explanation:

The company needs to manage encryption and decryption using keys retained outside of the AWS cloud. The AWS KMS external key store option (Option B) fulfills these requirements with minimal operational overhead. This option allows using an external key manager outside AWS while integrating with AWS KMS. The other options keep the keys within AWS.

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Question 43

Not Attempted

Storage



A solutions architect is designing a high-performance computing (HPC) environment on AWS. The workload requires numerous EC2 instances to concurrently access and process large datasets from a shared file system. The file system must offer sub-millisecond latency and support parallel read/write operations. After the initial processing, engineers need to access the processed data for post-processing analysis. Which storage solution best meets these requirements?

- ☐ Amazon Elastic File System (Amazon EFS) as a shared file system. Access the dataset from Amazon EFS.

Amazon EFS provides a scalable file system for use with AWS Cloud services and on-premises resources. While it's suitable for many workloads, it might not consistently meet the sub-millisecond latency requirement for HPC applications requiring high I/O performance.

- ☐ Mount an Amazon S3 bucket to serve as the shared file system. Perform post-processing directly from the S3 bucket.

Amazon S3 is object storage and not designed to be mounted as a file system for HPC workloads. The latency is significantly higher than 1ms, making it unsuitable for the stated requirements. Mounting S3 is generally done through FUSE, which introduces additional overhead and latency.

- ☒ Use Amazon FSx for Lustre as a shared file system. Link the file system to an Amazon S3 bucket for post-processing.

Amazon FSx for Lustre is specifically designed for HPC workloads. It provides a high-performance, parallel file system with sub-millisecond latencies and high throughput. Linking it to an S3 bucket allows for easy data transfer and post-processing access.

- ⊗ Configure AWS Resource Access Manager (RAM) to share an Amazon S3 bucket so that it can be mounted to all instances for processing and post-processing.

AWS RAM is used for sharing AWS resources, but it doesn't address the fundamental limitations of using S3 as a high-performance shared file system for HPC. Mounting S3 has high latency and doesn't support parallel processing effectively.

Overall Explanation:

Amazon FSx for Lustre is the optimal solution because it's designed for HPC workloads requiring high performance and low latency shared storage. It meets the sub-millisecond latency requirement, supports parallel access, and can be linked to S3 for post-processing needs. EFS does not guarantee the required latency for HPC, and S3 is not suitable as a mounted file system for HPC due to its inherent latency and limitations in parallel access.

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Question 44

Not Attempted

Networking



A gaming company is building an application with Voice over IP (VoIP) capabilities to serve users globally. The application requires high availability with automated failover across AWS Regions and minimized latency without relying on IP address caching on user devices. Which solution should a solutions architect implement to meet these requirements?

- ✔ Use AWS Global Accelerator with health checks.

AWS Global Accelerator uses static IP addresses (Anycast IPs) to provide a global entry point for applications. It routes traffic over the AWS global network to the optimal AWS endpoint based on health, geography, and routing policies. Health checks ensure traffic is routed only to healthy endpoints, providing high availability and automated failover. It's well-suited for non-HTTP use cases like VoIP (UDP) and minimizes latency.

- ⊗ Use Amazon Route 53 with a geolocation routing policy.

Amazon Route 53 with a geolocation routing policy routes traffic based on the user's geographic location. However, it relies on DNS caching, which the company wants to avoid.

- ⊗ Create an Amazon CloudFront distribution that includes multiple origins.

Amazon CloudFront is designed for caching and delivering static and dynamic web content, not for VoIP applications. It primarily handles HTTP/HTTPS traffic.

⊗ Create an Application Load Balancer that uses path-based routing.

An Application Load Balancer (ALB) with path-based routing routes traffic based on the URL path. It does not address the global availability, automated failover, or latency minimization requirements for VoIP traffic across multiple regions. ALBs are generally used within a single region.

Overall Explanation:

AWS Global Accelerator is the most suitable solution. It provides a global entry point with static IP addresses, routes traffic over the AWS global network for minimal latency, supports health checks for automated failover, and is well-suited for UDP-based applications like VoIP. It avoids reliance on IP address caching on user devices by utilizing Anycast IPs. Route 53 relies on DNS caching, CloudFront is not designed for VoIP, and ALB does not handle global distribution or UDP traffic effectively.

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Question 45

Not Attempted

Storage



A weather forecasting company requires a highly available, cloud storage solution with sub-millisecond latency to process hundreds of gigabytes of data. Their HPC environment will expand to the cloud, needing simultaneous access and processing of the entire dataset by thousands of compute instances. Which storage solution best fits these requirements?

⊗ Use Amazon FSx for Lustre scratch file systems.

Amazon FSx for Lustre scratch file systems are designed for temporary storage and do not offer the data persistence required.

✔ Use Amazon FSx for Lustre persistent file systems.

Amazon FSx for Lustre persistent file systems are suitable for HPC workloads that need high availability and sustained throughput, allowing thousands of compute instances to simultaneously access and process large datasets.

⊗ Use Amazon Elastic File System (Amazon EFS) with Bursting Throughput mode.

Amazon EFS with Bursting Throughput mode may not provide the sustained, sub-millisecond latency and high throughput required for HPC workloads processing large datasets.

⊗ Use Amazon Elastic File System (Amazon EFS) with Provisioned Throughput mode.

While Amazon EFS with Provisioned Throughput offers better performance than Bursting Throughput, it still may not match the performance and scalability of FSx for Lustre for demanding HPC workloads.

Overall Explanation:

Amazon FSx for Lustre persistent file systems are the best choice for high-performance computing (HPC) workloads that require high availability and the ability for thousands of compute instances to simultaneously access and process large datasets with low latency. Scratch file systems are not designed for persistent data. While EFS can be used, it doesn't provide the same level of performance as Lustre for this type of application. Therefore, only FSx for Lustre provides HA and high-performance

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Question 46

Not Attempted

Database



An e-commerce company is migrating their on-premises PostgreSQL database to Amazon RDS for PostgreSQL. The current on-premises database uses high IOPS Amazon Elastic Block Store (Amazon EBS) block storage, with daily peak I/O transactions not exceeding 15,000 IOPS. The company needs to provision disk IOPS performance independently of storage capacity in RDS and wants the MOST cost-effective solution. Which option meets these requirements?

⊗ Configure the General Purpose SSD (gp2) EBS volume storage type and provision 15,000 IOPS.

gp2's IOPS are tied to volume size and while it can reach 15,000 IOPS, it might not be the most cost-effective option, especially if IOPS is needed independent of storage capacity. Also, gp3 is cheaper than gp2.

⊗ Configure the Provisioned IOPS SSD (io1) EBS volume storage type and provision 15,000 IOPS.

io1 is designed for high-performance, IOPS-intensive workloads. However, it is more expensive than gp3, which can also meet the 15,000 IOPS requirement.

✓ Configure the General Purpose SSD (gp3) EBS volume storage type and provision 15,000 IOPS.

gp3 allows you to provision IOPS independently of storage capacity and is generally more cost-effective than io1 and gp2, especially for workloads within its performance limits. It is also cheaper than gp2 while providing the same IOPS.

⊗ Configure the EBS magnetic volume type to achieve maximum IOPS.

EBS magnetic volumes are the lowest-cost storage option, but they do not provide the consistent performance or high IOPS required for a database workload and are not suitable for this scenario. They also do not support provisioning IOPS.

Overall Explanation:

The MOST cost-effective solution is to use gp3. gp3 allows independent scaling of IOPS and storage, and can reach the required 15,000 IOPS. It's also cheaper than gp2 and io1 for this performance level. io1 is overkill and more expensive, while gp2 has IOPS tied to volume size. Magnetic volumes are not suitable for the required IOPS performance. Therefore, gp3 offers the best balance of cost and performance.

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Question 47

Not Attempted

Serverless



A hospital recently deployed a RESTful API with Amazon API Gateway and AWS Lambda. The hospital uses API Gateway and Lambda to upload reports that are in PDF format and JPEG format. The hospital needs to modify the Lambda code to identify protected health information (PHI) in the reports. Which solution will meet these requirements with the LEAST operational overhead?

⊗ Use existing Python libraries to extract the text from the reports and to identify the PHI from the extracted text.

Using existing Python libraries would require significant custom development and maintenance, leading to higher operational overhead.

⊗ Use Amazon Textract to extract the text from the reports. Use Amazon SageMaker to identify the PHI from the extracted text.

Using SageMaker for PHI identification introduces unnecessary complexity and overhead, as it requires model training and management.

- ✓ Use Amazon Textract to extract the text from the reports. Use Amazon Comprehend Medical to identify the PHI from the extracted text.

This is the most efficient solution. Textract extracts the text, and Comprehend Medical, a managed service, identifies PHI with minimal overhead.

- ✗ Use Amazon Rekognition to extract the text from the reports. Use Amazon Comprehend Medical to identify the PHI from the extracted text.

Rekognition is optimized for spatial text recognition in images, not document processing. Textract is more appropriate for extracting text from PDFs and documents.

Overall Explanation:

The optimal solution involves using Amazon Textract to extract text from the PDF and JPEG reports, followed by using Amazon Comprehend Medical to identify PHI. This approach minimizes operational overhead by leveraging managed services specifically designed for these tasks. Textract is designed for extracting text from documents, while Comprehend Medical is tailored to identify PHI in medical text, requiring minimal maintenance and eliminating the need for custom machine learning models or extensive custom code.

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Question 48

Not Attempted

Database



A company wants to migrate its on-premises Microsoft SQL Server Enterprise edition database to AWS. The company's online application uses the database to process transactions. The data analysis team uses the same production database to run reports for analytical processing. The company wants to reduce operational overhead by moving to managed services wherever possible. Which solution will meet these requirements with the LEAST operational overhead?

- ✓ Migrate to Amazon RDS for Microsoft SQL Server. Use read replicas for reporting purposes.

RDS provides managed services, reducing operational overhead. Read replicas are easily created and asynchronously replicated, suitable for reporting with a slight delay.

- ✗ Migrate to Microsoft SQL Server on Amazon EC2. Use Always On read replicas for reporting purposes.

While possible, managing SQL Server on EC2 involves more operational overhead compared to RDS.

- ⊗ Migrate to Amazon DynamoDB. Use DynamoDB on-demand replicas for reporting purposes.

DynamoDB is a NoSQL database and not compatible with the existing Microsoft SQL Server database.

- ⊗ Migrate to Amazon Aurora MySQL. Use Aurora read replicas for reporting purposes.

Migrating from SQL Server to MySQL involves significant changes and potential loss of SQL Server functionalities.

Overall Explanation:

Amazon RDS for Microsoft SQL Server with read replicas offers the least operational overhead as it is a managed service. The read replicas can be used for reporting purposes without impacting the performance of the primary database. Migrating to a different database engine like DynamoDB or Aurora MySQL would require significant application changes, and running SQL Server on EC2 increases the operational burden.

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Question 49

Not Attempted

Storage



A company stores a large volume of image files in an Amazon S3 bucket. The images must be readily available for the first 180 days. For the subsequent 180 days, the images are accessed infrequently. After 360 days, the images need to be archived but must be available instantly upon request. After 5 years, only auditors can access the images, who require retrieval within 12 hours. Data loss is unacceptable during the entire process. A developer intends to use S3 Standard storage for the initial 180 days and needs to configure an S3 Lifecycle rule to manage the storage tiers effectively. Which solution will meet these requirements MOST cost-effectively?

- ⊗ Transition the objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 180 days, S3 Glacier Instant Retrieval after 360 days, and S3 Glacier Deep Archive after 5 years.

S3 One Zone-IA does not provide the required high availability to prevent data loss. It also cannot transition to S3 Glacier Instant Retrieval directly.

- ⊗ Transition the objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 180 days, S3 Glacier Flexible Retrieval after 360 days, and S3 Glacier Deep Archive after 5 years.

S3 One Zone-IA does not provide the required high availability to prevent data loss. S3 Glacier Flexible Retrieval does not provide instant access, which is a requirement.

- ✓ Transition the objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 180 days, S3 Glacier Instant Retrieval after 360 days, and S3 Glacier Deep Archive after 5 years.

This option meets the requirements: S3 Standard-IA provides high availability and cost savings for infrequently accessed data, S3 Glacier Instant Retrieval provides instant access to archived images, and S3 Glacier Deep Archive provides the lowest cost archival storage with acceptable retrieval times for auditors.

- ⊗ Transition the objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 180 days, S3 Glacier Flexible Retrieval after 360 days, and S3 Glacier Deep Archive after 5 years.

S3 Glacier Flexible Retrieval does not provide instant access, which is a requirement.

Overall Explanation:

The most cost-effective solution is to use S3 Standard-Infrequent Access (S3 Standard-IA) after 180 days for infrequently accessed data. After 360 days, transitioning to S3 Glacier Instant Retrieval ensures that the archived images are available instantly upon request. Finally, transitioning to S3 Glacier Deep Archive after 5 years provides the lowest cost archival storage for infrequently accessed audit data with retrieval times within the required 12-hour window. This combination of storage classes satisfies all requirements while optimizing cost.

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Question 50

Not Attempted

Compute



A company runs a critical data workload for 6 hours daily and cannot afford data loss during processing. They're designing an Amazon EMR cluster configuration. Which solution is the MOST cost-effective while ensuring no data loss?

- ⊗ Configure a long-running cluster with On-Demand Instances for primary and core nodes, and Spot Instances for task nodes.

A long-running cluster runs continuously, which is less cost-effective for a 6-hour workload. While using Spot Instances for task nodes reduces costs, the long-running nature isn't ideal.

- ✔ Configure a transient cluster with On-Demand Instances for primary and core nodes, and Spot Instances for task nodes.

This is the most cost-effective solution. A transient cluster runs only for the duration of the workload. Using On-Demand Instances for primary and core nodes ensures data consistency, while Spot Instances for task nodes minimizes cost.

- ✘ Configure a transient cluster with an On-Demand Instance for the primary node, and Spot Instances for core and task nodes.

Using Spot Instances for core nodes can lead to data loss if the Spot Instances are interrupted. Core nodes are crucial for data storage and processing in EMR.

- ✘ Configure a long-running cluster with an On-Demand Instance for the primary node, and Spot Instances for core and task nodes.

A long-running cluster is not cost-effective for a 6-hour daily workload. Also, running core nodes on spot instances risks data loss.

Overall Explanation:

The best solution is a transient cluster with On-Demand primary and core nodes, and Spot Instances for task nodes. Transient clusters only run when needed, saving costs. On-Demand for primary and core nodes protects against data loss, while Spot Instances for task nodes provide further cost savings. Using Spot instances for core nodes is risky as it can lead to data loss if spot instances are terminated.

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Question 51

Not Attempted

Automation



A company maintains an Amazon RDS database that maps users to cost centers. The company has accounts in an organization in AWS Organizations. The company needs a solution that will tag all resources that are created in a specific AWS account in the organization. The solution must tag each resource with the cost center ID of the user who created the resource. Which solution will meet these requirements?

- ⊗ Move the specific AWS account to a new organizational unit (OU) in Organizations from the management account. Create a service control policy (SCP) that requires all existing resources to have the correct cost center tag before the resources are created. Apply the SCP to the new OU.

SCP cannot look up the cost center ID of the user who created the resource. SCPs are used to enforce permissions and restrictions across accounts in an organization and cannot be used to dynamically tag resources based on user information stored in an RDS database. Also, SCPs cannot be used to tag existing resources

- ✔ Create an AWS Lambda function to tag the resources after the Lambda function looks up the appropriate cost center from the RDS database. Configure an Amazon EventBridge rule that reacts to AWS CloudTrail events to invoke the Lambda function.

This solution utilizes AWS Lambda and Amazon EventBridge to automate the tagging process based on information from the RDS database and CloudTrail events, tagging resources as they are created.

- ⊗ Create an AWS CloudFormation stack to deploy an AWS Lambda function. Configure the Lambda function to look up the appropriate cost center from the RDS database and to tag resources. Create an Amazon EventBridge scheduled rule to invoke the CloudFormation stack.

A scheduled rule will not be triggered when the resource is created. This means that resources would temporarily exist without being tagged.

- ⊗ Create an AWS Lambda function to tag the resources with a default value. Configure an Amazon EventBridge rule that reacts to AWS CloudTrail events to invoke the Lambda function when a resource is missing the cost center tag.

This tags the resources with a default value which is not required.

Overall Explanation:

The best solution is to use a Lambda function triggered by an EventBridge rule based on CloudTrail events. The Lambda function looks up the cost center ID from the RDS database and tags the resource. This allows resources to be dynamically tagged with the appropriate cost center ID upon creation. Using a default value or SCPs will not meet the requirements since the solution has to look up the cost center id of the user from the RDS database. Scheduled rules will not trigger the lambda upon creation of the resource.

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Question 52

Not Attempted

Compute



A company recently migrated its web application to the AWS Cloud. The company uses an Amazon EC2 instance to run multiple processes to host the application. The processes include an Apache web server that serves static content. The Apache web server makes requests to a PHP application that uses a local Redis server for user sessions. The company wants to redesign the architecture to be highly available and to use AWS managed solutions. Which solution will meet these requirements?

- ⊗ Use AWS Elastic Beanstalk to host the static content and the PHP application. Configure Elastic Beanstalk to deploy its EC2 instance into a public subnet. Assign a public IP address.

Elastic Beanstalk, while managed, deploying EC2 instances into a public subnet with a public IP is not the most secure or scalable solution, and doesn't fully leverage managed services for static content.

- ⊗ Use AWS Lambda to host the static content and the PHP application. Use an Amazon API Gateway REST API to proxy requests to the Lambda function. Set the API Gateway CORS configuration to respond to the domain name. Configure Amazon ElastiCache for Redis to handle session information.

While Lambda can host applications, it's not ideal for serving static content directly or running a complex PHP application with session management. It also doesn't naturally address the need for high availability without significant configuration overhead.

- ⊗ Keep the backend code on the EC2 instance. Create an Amazon ElastiCache for Redis cluster that has Multi-AZ enabled. Configure the ElastiCache for Redis cluster in cluster mode. Copy the frontend resources to Amazon S3. Configure the backend code to reference the EC2 instance.

Keeping the backend code on a single EC2 instance doesn't address the high availability requirement. While ElastiCache Multi-AZ and S3 for static content are good, the core application remains a single point of failure.

- ✔ Configure an Amazon CloudFront distribution with an Amazon S3 endpoint to an S3 bucket that is configured to host the static content. Configure an Application Load Balancer that targets an Amazon Elastic Container Service (Amazon ECS) service that runs AWS Fargate tasks for the PHP application. Configure the PHP application to use an Amazon ElastiCache for Redis cluster that runs in multiple Availability Zones.

This is the best solution. It uses CloudFront and S3 for highly available static content delivery, ECS with Fargate for a scalable and managed PHP application, and ElastiCache for Redis in Multi-AZ for

highly available session management. All components leverage managed services, promoting high availability.

Overall Explanation:

The correct answer is D. It utilizes CloudFront and S3 for static content, ECS with Fargate for the PHP application's backend, and ElastiCache (Redis) with Multi-AZ for session management. This combination provides high availability and leverages AWS managed services effectively. Options A and B are less suitable because they either don't scale well or aren't appropriate for the type of workload. Option C doesn't address the single point of failure with the EC2 instance.

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Question 53

Not Attempted

Networking



A company hosts a web application on Amazon EC2 instances within an Auto Scaling group behind a target group. The application requires session affinity (sticky sessions) to enhance user experience. The application must be publicly accessible over the internet, secured by a WAF, and maintain sticky sessions.

- ☒ Create a public Network Load Balancer. Specify the application target group.
Network Load Balancers do not support session affinity (sticky sessions).
- ☒ Create a Gateway Load Balancer. Specify the application target group.
Gateway Load Balancers do not support session affinity (sticky sessions). They are typically used for third-party appliances.
- ☒ Create a public Application Load Balancer. Specify the application target group.
Application Load Balancers support session affinity (sticky sessions) based on cookies and can be publicly accessible.
- ☒ Create a second target group. Add Elastic IP addresses to the EC2 instances.
Using Elastic IPs on EC2 instances negates the purpose of sticky sessions and is not a best practice for load balancing.
- ☒ Create a web ACL in AWS WAF. Associate the web ACL with the endpoint.

AWS WAF provides web application firewall capabilities, and associating a web ACL with the endpoint (Application Load Balancer) adds security.

Overall Explanation:

To meet the requirements, an Application Load Balancer (ALB) should be used to provide session affinity (sticky sessions) and make the application publicly accessible. AWS WAF should be implemented with a web ACL associated with the ALB to provide additional security. NLB and GLB cannot handle sticky sessions. Elastic IP will negate sticky sessions and this combination won't work.

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Question 54

Not Attempted

Storage



A company operates a website hosting images of historical events. Users need to search and view these images, primarily based on the year of the event depicted. On average, each image is requested only once or twice per year. The company requires a highly available and cost-effective solution for storing and delivering these images directly to users. Which solution meets these requirements MOST cost-effectively?

- ☐ Store images in Amazon Elastic Block Store (Amazon EBS). Use a web server that runs on Amazon EC2.

EBS is not ideal for infrequently accessed data and requires a running EC2 instance, making it less cost-effective and potentially impacting availability if the EC2 instance fails.

- ☐ Store images in Amazon Elastic File System (Amazon EFS). Use a web server that runs on Amazon EC2.

EFS is designed for shared file storage and is more expensive than object storage like S3, making it less cost-effective for this use case. It also requires a running EC2 instance.

- ☐ Store images in Amazon S3 Standard. Use S3 Standard to directly deliver images by using a static website.

S3 Standard is suitable for frequently accessed data. Given that images are only accessed once or twice a year, this is not the most cost-effective option due to higher storage costs.

- ☒ Store images in Amazon S3 Standard-Infrequent Access (S3 Standard-IA). Use S3 Standard-IA to directly deliver images by using a static website.

S3 Standard-IA is designed for infrequently accessed data, providing lower storage costs compared to S3 Standard while maintaining high availability and durability. Direct delivery via S3 further reduces operational overhead and cost.

Overall Explanation:

S3 Standard-IA offers the most cost-effective solution by providing lower storage costs for infrequently accessed data while maintaining high availability. Utilizing S3 for direct delivery eliminates the need for a web server, further reducing costs and operational complexity. EBS and EFS are not cost-effective. S3 Standard is for frequently accessed data, not infrequent access.

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Question 55

Not Attempted

Networking



A company has multiple AWS accounts in an organization in AWS Organizations that different business units use. The company has multiple offices around the world. The company needs to update security group rules to allow new office CIDR ranges or to remove old CIDR ranges across the organization. The company wants to centralize the management of security group rules to minimize the administrative overhead that updating CIDR ranges requires. Which solution will meet these requirements MOST cost-effectively?

- ☐ Create VPC security groups in the organization's management account. Update the security groups when a CIDR range update is necessary.
Creating security groups in the management account doesn't address the need to easily propagate changes across multiple accounts or centralize CIDR management.
- ☒ Create a VPC customer managed prefix list that contains the list of CIDRs. Use AWS Resource Access Manager (AWS RAM) to share the prefix list across the organization. Use the prefix list in the security groups across the organization.
This is the most cost-effective solution. A prefix list allows for centralized management of CIDR ranges. AWS RAM allows for sharing the prefix list across the organization. Updating the prefix list updates all security groups using it.
- ☐ Create an AWS managed prefix list. Use an AWS Security Hub policy to enforce the security group update across the organization. Use an AWS Lambda function to update the prefix list automatically when the CIDR ranges change.

AWS Security Hub policies are not designed to enforce security group updates. While a Lambda function could automate prefix list updates, it adds complexity and cost compared to option B.

- ⊗ Create security groups in a central administrative AWS account. Create an AWS Firewall Manager common security group policy for the whole organization. Select the previously created security groups as primary groups in the policy.

AWS Firewall Manager manages security groups but isn't as suitable for simple CIDR range updates as a prefix list. Firewall Manager has more overhead for this specific use case.

Overall Explanation:

The most cost-effective solution is to create a VPC customer managed prefix list, share it across the organization using AWS RAM, and reference it in security groups. This centralizes CIDR management, reduces administrative overhead, and ensures consistent security group rules across all accounts. A managed prefix list is a set of one or more CIDR blocks. You can use prefix lists to make it easier to configure and maintain your security groups and route tables. You can create a prefix list from the IP addresses that you frequently use, and reference them as a set in security group rules and routes instead of referencing them individually. If you scale your network and need to allow traffic from another CIDR block, you can update the relevant prefix list and all security groups that use the prefix list are updated. You can also use managed prefix lists with other AWS accounts using Resource Access Manager (RAM).

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Question 56

Not Attempted

Storage



A company is migrating its on-premises, latency-sensitive, high-performance computing (HPC) workloads to the AWS Cloud. The HPC workloads currently utilize an on-premises network-attached storage (NAS) system providing file shares. The company requires both NFS and SMB multi-protocol access to the file system in AWS with the LEAST possible latency. Which of the following solutions meet these requirements?

- ✔ Deploy compute optimized EC2 instances into a cluster placement group.

Deploying EC2 instances into a cluster placement group ensures the lowest possible latency between the instances, critical for latency-sensitive HPC workloads.

- ☐ Deploy compute optimized EC2 instances into a partition placement group.
A partition placement group focuses on fault tolerance and spreading instances across partitions, not necessarily minimizing latency like a cluster placement group does.
- ☐ Attach the EC2 instances to an Amazon FSx for Lustre file system.
Amazon FSx for Lustre is a high-performance file system, but it does not natively support both NFS and SMB protocols. It primarily focuses on POSIX compliance.
- ☐ Attach the EC2 instances to an Amazon FSx for OpenZFS file system.
Amazon FSx for OpenZFS provides NFS support, but less optimized for combined NFS/SMB access and potentially higher latency compared to ONTAP for this specific scenario.
- ☒ Attach the EC2 instances to an Amazon FSx for NetApp ONTAP file system.
Amazon FSx for NetApp ONTAP provides native support for both NFS and SMB protocols, crucial for the multi-protocol access requirement.

Overall Explanation:

The optimal solution for migrating latency-sensitive HPC workloads with NFS and SMB multi-protocol access to AWS with the least latency involves deploying compute-optimized EC2 instances within a cluster placement group (for minimal network latency) and attaching them to an Amazon FSx for NetApp ONTAP file system (for NFS and SMB protocol support). Lustre does not support SMB and OpenZFS is less suitable than ONTAP in a heterogeneous setup with NFS and SMB. Placement group has been set to cluster to optimize network latency for HPC workloads.

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Question 57

Not Attempted

Networking



A company is relocating its data center and needs to securely transfer 50 TB of data to AWS within 2 weeks. The existing data center has a Site-to-Site VPN connection to AWS that is 90% utilized. Which AWS service should a solutions architect use to meet these requirements?

- ☐ AWS DataSync with a VPC endpoint
DataSync is suitable for online data transfer, but given the 90% VPN utilization, it may not be the most efficient solution for transferring 50 TB within 2 weeks.

⊗ AWS Direct Connect

While Direct Connect provides a dedicated network connection, setting it up can take longer than 2 weeks. Therefore, it's not the optimal solution given the time constraint.

✓ AWS Snowball Edge Storage Optimized

AWS Snowball Edge Storage Optimized is designed for large-scale data transfers when network bandwidth is limited. Given the 50 TB data volume and the existing VPN's 90% utilization, it's the most suitable option to meet the 2-week deadline.

⊗ AWS Storage Gateway

AWS Storage Gateway requires continuous network connectivity and is not ideal for transferring a large amount of data in a short time frame when the existing network connection is already highly utilized.

Overall Explanation:

AWS Snowball Edge Storage Optimized is the best option because it provides a secure, offline method for transferring large amounts of data to AWS when network bandwidth is limited or unavailable. Given the 50 TB data volume, the 2-week timeframe, and the existing VPN's 90% utilization, Snowball Edge is the most efficient solution. DataSync would be limited by the already utilized VPN connection. Direct Connect takes longer than 2 weeks to provision. Storage Gateway relies on network bandwidth which is limited.

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Question 58

Not Attempted

Storage



A company's application generates numerous 5 MB files, stored in Amazon S3. Company policy mandates a 4-year retention period before deletion. Immediate data accessibility is critical as the files contain irreplaceable business data. The files are accessed frequently in the first 30 days but rarely afterward. Which storage lifecycle solution is MOST cost-effective?

⊗ Create an S3 bucket lifecycle policy to move files from S3 Standard to S3 Glacier 30 days from object creation. Delete the files 4 years after object creation.

S3 Glacier, without specifying Instant Retrieval, generally has longer retrieval times, potentially violating the 'Immediate accessibility is always required' constraint.

- ⊗ Create an S3 bucket lifecycle policy to move files from S3 Standard to S3 One Zone-Infrequent Access (S3 One Zone-IA) 30 days from object creation. Delete the files 4 years after object creation.

While S3 One Zone-IA is cost-effective, storing critical, irreplaceable business data in a single Availability Zone poses a risk of data loss in case of AZ failure. The question does not explicitly rule out high availability.

- ✔ Create an S3 bucket lifecycle policy to move files from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) 30 days from object creation. Delete the files 4 years after object creation.

S3 Standard-IA provides immediate accessibility with lower storage costs for infrequently accessed data compared to S3 Standard, meeting both the accessibility and cost requirements. The files are frequently accessed in the first 30 days before being transferred.

- ⊗ Create an S3 bucket lifecycle policy to move files from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) 30 days from object creation. Move the files to S3 Glacier 4 years after object creation.

This option moves files to S3 Standard IA after 30 days, but then suggests moving them to S3 Glacier 4 years after object creation. The question asks for the files to be deleted 4 years after creation not archived.

Overall Explanation:

S3 Standard-IA is the most suitable option. It maintains immediate accessibility, reduces storage costs for infrequently accessed data after 30 days, and aligns with the 4-year retention policy before deletion. S3 Glacier is not appropriate because it doesn't fulfill immediate accessibility needs. S3 One Zone-IA is also not appropriate as it stores data in one availability zone which could lead to data loss for critical business data.

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Question 59

Not Attempted

Compute



A company hosts an application on Amazon EC2 On-Demand Instances in an Auto Scaling group. Application peak hours occur at the same time each day. Application users report slow application performance at the start of peak hours. The application performs normally 2-3 hours after peak hours begin. The company wants to ensure that the application works properly at the start of peak hours. Which solution will meet these requirements?

- ☐ **Configure an Application Load Balancer to distribute traffic properly to the instances.**
Configuring an Application Load Balancer distributes traffic but doesn't address the initial performance issues during peak hours.
- ☐ **Configure a dynamic scaling policy for the Auto Scaling group to launch new instances based on memory utilization.**
Dynamic scaling based on memory utilization will react to increased memory usage, but won't proactively address the performance issues at the **start** of peak hours.
- ☐ **Configure a dynamic scaling policy for the Auto Scaling group to launch new instances based on CPU utilization.**
Dynamic scaling based on CPU utilization will react to increased CPU usage, but won't proactively address the performance issues at the **start** of peak hours.
- ☒ **Configure a scheduled scaling policy for the Auto Scaling group to launch new instances before peak hours.**
A scheduled scaling policy allows you to proactively launch new instances before the predictable peak hours, ensuring capacity is available to handle the load from the start.

Overall Explanation:

Since the application peak hours occur at the same time each day (predictable), a scheduled scaling policy is the best approach. It allows the Auto Scaling group to launch new instances **before** the peak hours begin, ensuring that sufficient capacity is available to handle the increased load from the start and preventing the slow performance reported by users. Dynamic scaling would react **after** the performance issue starts.

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Question 60

Not Attempted

Database



A company runs applications on AWS that connect to the company's Amazon RDS database. The applications scale on weekends and at peak times of the year. The company wants to scale the database connections more effectively for its applications that connect to the database, with the LEAST operational overhead.

- ☐ **Use Amazon DynamoDB with connection pooling with a target group configuration for the database. Change the applications to use the DynamoDB endpoint.**

DynamoDB is a NoSQL database and is not compatible with applications designed for an RDS database.

- ✔ Use Amazon RDS Proxy with a target group for the database. Change the applications to use the RDS Proxy endpoint.

Amazon RDS Proxy is a fully managed, highly available database proxy designed to improve database efficiency and application scalability by pooling and sharing connections.

- ✘ Use a custom proxy that runs on Amazon EC2 as an intermediary to the database. Change the applications to use the custom proxy endpoint.

Implementing a custom proxy on EC2 requires significant operational overhead for management, scaling, and high availability.

- ✘ Use an AWS Lambda function to provide connection pooling with a target group configuration for the database. Change the applications to use the Lambda function.

Using Lambda for connection pooling is impractical and inefficient due to Lambda's invocation model and limitations.

Overall Explanation:

Amazon RDS Proxy is the most suitable solution because it's a fully managed service that specifically addresses connection pooling for RDS databases with the least operational overhead. It allows applications to pool and share database connections, improving efficiency and scalability. DynamoDB is a NoSQL database and not compatible. A custom proxy on EC2 and Lambda involve greater operational overhead.

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Question 61

Not Attempted

Storage



A company uses AWS Cost Explorer to monitor its AWS costs. The company notices that Amazon Elastic Block Store (Amazon EBS) storage and snapshot costs increase every month. However, the company does not purchase additional EBS storage every month. The company wants to optimize monthly costs for its current storage usage. Which solution will meet these requirements with the LEAST operational overhead?

- ✘ Use logs in Amazon CloudWatch Logs to monitor the storage utilization of Amazon EBS. Use Amazon EBS Elastic Volumes to reduce the size of the EBS volumes.

Monitoring EBS utilization can help reduce costs by resizing volumes, but it doesn't directly address snapshot costs. Requires manual effort and lacks automated lifecycle management.

- ⊗ Use a custom script to monitor space usage. Use Amazon EBS Elastic Volumes to reduce the size of the EBS volumes.

Using a custom script adds significant operational overhead for monitoring and management, particularly as the environment scales. Doesn't address snapshot management.

- ⊗ Delete all expired and unused snapshots to reduce snapshot costs.

While deleting expired and unused snapshots will reduce costs, it's a one-time effort and does not provide an ongoing solution for managing snapshots. Also, identifying expired/unused snapshots manually is operationally intensive.

- ✓ Delete all nonessential snapshots. Use Amazon Data Lifecycle Manager to create and manage the snapshots according to the company's snapshot policy requirements.

This option provides a sustainable solution for optimizing snapshot costs by deleting non-essential snapshots and using Amazon Data Lifecycle Manager (DLM) to automate the creation and deletion of snapshots based on defined policies, minimizing operational overhead.

Overall Explanation:

The best solution is to delete nonessential snapshots and use Amazon Data Lifecycle Manager (DLM). DLM automates the creation and deletion of snapshots based on defined policies, reducing operational overhead. Options A and B focus on EBS volume resizing but do not address snapshot costs effectively and require manual intervention. Option C only addresses a one-time cleanup of snapshots, not ongoing management.

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Question 62

Not Attempted

Security



A company is developing a new application on AWS. The application consists of an Amazon Elastic Container Service (Amazon ECS) cluster, an Amazon S3 bucket that contains assets for the application, and an Amazon RDS for MySQL database that contains sensitive information. The company requires that only the ECS cluster can access the data in the RDS for MySQL database and the data in the S3 bucket. Which solution meets these requirements?

- ✓ Create a new AWS Key Management Service (AWS KMS) customer managed key to encrypt both the S3 bucket and the RDS for MySQL database. Ensure that the KMS key policy includes encrypt and decrypt permissions for the ECS task execution role.

This option ensures that only the ECS task execution role can encrypt and decrypt data in S3 and RDS through the KMS key policy, thus restricting access.

- ✗ Create an AWS Key Management Service (AWS KMS) AWS managed key to encrypt both the S3 bucket and the RDS for MySQL database. Ensure that the S3 bucket policy specifies the ECS task execution role as a user.

Using an AWS managed key does not provide sufficient control over access, and specifying the ECS task execution role in the S3 bucket policy alone does not restrict RDS access.

- ✗ Create an S3 bucket policy that restricts bucket access to the ECS task execution role. Create a VPC endpoint for Amazon RDS for MySQL. Update the RDS for MySQL security group to allow access from only the subnets that the ECS cluster will generate tasks in.

While this restricts S3 access, it allows RDS access from the entire subnet where ECS tasks run, potentially allowing other resources to access the database.

- ✗ Create a VPC endpoint for Amazon RDS for MySQL. Update the RDS for MySQL security group to allow access from only the subnets that the ECS cluster will generate tasks in. Create a VPC endpoint for Amazon S3. Update the S3 bucket policy to allow access from only the S3 VPC endpoint.

Similar to option C, this approach allows RDS access from the entire ECS subnet and S3 access from the VPC endpoint, which may be accessed by other resources.

Overall Explanation:

Option A is the most secure because it uses a KMS customer managed key, ensuring only the ECS task execution role can encrypt and decrypt the data in both the S3 bucket and the RDS for MySQL database. This satisfies the requirement of restricting access to only the ECS cluster. Options C and D use network controls via VPC endpoints, but do not granularly control access to the data itself based on the ECS task execution role. Option B doesn't properly restrict access to the data.

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Question 63

Not Attempted

Compute



A company's on-premises web application experiences latency issues during peak hours, occurring twice monthly, with CPU utilization spiking to 10 times the normal amount at the onset of these issues. The company plans to migrate the application to AWS using Elastic Beanstalk to improve latency and automatically scale with demand. Which solution best meets these requirements?

- ☒ Configure an Elastic Beanstalk environment to use burstable performance instances in unlimited mode. Configure the environment to scale based on requests.

Burstable performance instances with unlimited mode allow for sustained high CPU utilization when needed, accommodating the sudden CPU spikes. Scaling based on requests aligns with the need to automatically scale when application demand increases.

- ☐ Configure an Elastic Beanstalk environment to use compute optimized instances. Configure the environment to scale based on requests.

Compute-optimized instances might be an overkill for the average CPU utilization, and won't solve 10x CPU issues at the start of the latency. Compute Optimized instances are more suited for consistently high CPU workloads.

- ☐ Configure an Elastic Beanstalk environment to use compute optimized instances. Configure the environment to scale on a schedule.

Scaling on a schedule is inappropriate because the application demand increases at unpredictable times and will not dynamically adjust to changes in demand.

- ☐ Configure an Elastic Beanstalk environment to use burstable performance instances in unlimited mode. Configure the environment to scale on predictive metrics.

While burstable performance instances are suitable, Elastic Beanstalk does not natively support scaling based on predictive metrics. Moreover, in a migration scenario, there is unlikely to be sufficient historical data available initially to implement predictive scaling.

Overall Explanation:

Option A is the most suitable because it leverages burstable performance instances in unlimited mode to handle the sudden CPU spikes and configures scaling based on requests, aligning with the requirement for automatic scaling based on demand. Options B and C suggest Compute Optimized instances which are an overkill and won't help much at the start of the latency. Scaling on a schedule is also not ideal. Scaling on predictive metrics (option D) is not a feature of elastic beanstalk.

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Question 64

Not Attempted

Automation



A company with a global customer base seeks to automate the securing of its systems and network infrastructure. The security team requires a solution to track and audit all incremental infrastructure changes automatically. Which of the following solutions BEST satisfies these requirements?

- ☐ Use AWS Organizations to set up the infrastructure. Use AWS Config to track changes.
AWS Organizations helps manage multiple AWS accounts, but it doesn't directly provide infrastructure automation or detailed change tracking. AWS Config tracks changes but Organizations is not needed.
- ☒ Use AWS CloudFormation to set up the infrastructure. Use AWS Config to track changes.
AWS CloudFormation allows defining infrastructure as code for repeatable and automated deployments. AWS Config provides the necessary tracking and auditing capabilities for infrastructure changes.
- ☐ Use AWS Organizations to set up the infrastructure. Use AWS Service Catalog to track changes.
AWS Organizations is not directly related to infrastructure automation and change tracking. AWS Service Catalog focuses on managing approved IT services but does not inherently provide detailed change tracking for all infrastructure components.
- ☐ Use AWS CloudFormation to set up the infrastructure. Use AWS Service Catalog to track changes.
AWS CloudFormation provides infrastructure automation but AWS Service Catalog is not a change tracking tool. It manages approved IT services and templates.

Overall Explanation:

The best solution is to use AWS CloudFormation for infrastructure as code and AWS Config for change tracking and auditing. CloudFormation enables consistent and automated deployments, while AWS Config provides visibility into changes and configuration details, addressing both requirements. Organizations is more for managing multi-account setups, and Service Catalog is more for service management.

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Question 65

Not Attempted

Database



A startup company hosts a website for its customers on an Amazon EC2 instance. The website consists of a stateless Python application and a MySQL database. The website serves only a small amount of traffic. The company is concerned about the reliability of the instance and needs to migrate to a highly available architecture. The company cannot modify the application code. Which combination of actions should a solutions architect take to achieve high availability for the website? (Choose two.)

- ☐ Provision an internet gateway in each Availability Zone in use.
While internet gateways are necessary for internet access, provisioning them in each Availability Zone does not directly contribute to high availability in the context of this scenario.
- ☒ Migrate the database to an Amazon RDS for MySQL Multi-AZ DB instance.
Migrating the database to RDS for MySQL with Multi-AZ configuration provides high availability by synchronously replicating the data to a standby instance in a different Availability Zone. If the primary instance fails, RDS automatically fails over to the standby instance, minimizing downtime.
- ☐ Migrate the database to Amazon DynamoDB, and enable DynamoDB auto scaling.
Migrating to DynamoDB would require application code changes, which the company wants to avoid. Also, DynamoDB is a NoSQL database and cannot be directly substituted for MySQL without significant application refactoring.
- ☐ Use AWS DataSync to synchronize the database data across multiple EC2 instances.
Using DataSync for database replication is not a suitable approach for high availability. DataSync is designed for periodic data transfers and not for real-time synchronous replication required for database failover.
- ☒ Create an Application Load Balancer to distribute traffic to an Auto Scaling group of EC2 instances that are distributed across two Availability Zones.
An Application Load Balancer (ALB) distributing traffic across an Auto Scaling group of EC2 instances in multiple Availability Zones ensures high availability for the application tier. If one instance or Availability Zone fails, the ALB automatically routes traffic to healthy instances in other Availability Zones.

Overall Explanation:

To achieve high availability, the database should be migrated to Amazon RDS for MySQL Multi-AZ DB instance for database redundancy and failover capabilities. Additionally, an Application Load Balancer should be used to distribute traffic to an Auto Scaling group of EC2 instances across multiple Availability Zones, ensuring application availability in case of instance or Availability Zone failures. Migrating the database to DynamoDB requires code changes, and DataSync is not suited for real-time replication. Provisioning an internet gateway per AZ doesn't directly address HA for the application or database in this context.

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